7-7b

1. Let f be a function that is differentiable throughout its domain and that has the following properties.

(i) $f\,(x+y)=\frac{f(x)+f(y)}{1-f(x)f(y)}$ for all real numbers x, y, and $x+y$ in the domain of f

(ii) $\lim\limits_{h\to 0} f\,(h)=0$

(iii) $\lim\limits_{h\to 0}\frac{f(h)}{h}=1$

(a) Show that $f(0)=0$.

(b) Use the definition of the derivative to show that $f'(x)=1+[f(x)]^2$. Indicate clearly where properties (i), (ii) and (iii) are used.

(c) Find $f(x)$ by solving the differential equation in part (b).

Part A

1 point is earned for to show $f(0)=0$

Method 1:

f is continuous at 0, so $f(0)=\lim\limits_{x\to 0} f(x)=0$

Method 2:

$$f(0)=f(0+0)=\frac{f(0)+f(0)}{1-f(0)f(0)}$$
$$f(0)\left(1-[f(0)]^2\right)=2f(0)$$
$$f(0)\left(-1-[f(0)]^2\right)=0$$
$$f(0)=0$$

0	1

The student response earns all of the following points:

1 point is earned for to show $f(0)=0$

Method 1:

f is continuous at 0, so $f(0)=\lim\limits_{x\to 0} f(x)=0$

Method 2:

$$f(0)=f(0+0)=\frac{f(0)+f(0)}{1-f(0)f(0)}$$
$$f(0)\left(1-[f(0)]^2\right)=2f(0)$$
$$f(0)\left(-1-[f(0)]^2\right)=0$$
$$f(0)=0$$

Part B

1 point is earned for definition of $f'(x)$

1 point is earned for clearly uses (i)

1 point is earned for clearly uses (ii)

7-7b

1 point is earned for clearly uses (iii)

1 point is earned for algebraic manipulation to answer

$$
\begin{aligned}
f'(x) &= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0}\frac{\frac{f(x)+f(h)}{1-f(x)f(h)}-f(x)}{h} \qquad \text{[By (i)]} \\
&= \lim_{h\to 0}\left[\frac{f(h)}{h}\cdot\frac{1+[f(x)]^2}{1-f(x)f(h)}\right] \\
&= 1\cdot\frac{1+[f(x)]^2}{1-f(x)\cdot 0} \qquad \text{[By (iii) \& (ii)]} \\
&= 1+[f(x)]^2
\end{aligned}
$$

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for definition of $f'(x)$

1 point is earned for clearly uses (i)

1 point is earned for clearly uses (ii)

1 point is earned for clearly uses (iii)

1 point is earned for algebraic manipulation to answer

$$
\begin{aligned}
f'(x) &= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0}\frac{\frac{f(x)+f(h)}{1-f(x)f(h)}-f(x)}{h} \qquad \text{[By (i)]} \\
&= \lim_{h\to 0}\left[\frac{f(h)}{h}\cdot\frac{1+[f(x)]^2}{1-f(x)f(h)}\right] \\
&= 1\cdot\frac{1+[f(x)]^2}{1-f(x)\cdot 0} \qquad \text{[By (iii) \& (ii)]} \\
&= 1+[f(x)]^2
\end{aligned}
$$

Part C

1 point is earned if separates variables

1 point is earned for integration, with C

1 point is earned for c evaluated

<u>Method 1:</u>

Let $y = f(x)$; $\dfrac{dy}{dx} = 1 + y^2$

$$\frac{dy}{1+y^2} = dx$$

$$\tan^{-1} y = x + C$$

$$y = \tan(x + C)$$

$$f(0) = 0 \Rightarrow C = 0 \ \text{[or } C = n\pi, n \in \mathbb{Z}\text{]}$$

$$f(x) = \tan x \ \text{[or } f(x) = \tan(x + n\pi)\text{]}$$

7-7b

or

Method 2:

Guess that $f(x) = \tan x$

$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$
$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique $f(x) = \tan x$ is the solution.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned if separates variables

1 point is earned for integration, with C

1 point is earned for c evaluated

Method 1:

Let $y = f(x)$; $\dfrac{dy}{dx} = 1 + y^2$
$$\frac{dy}{1+y^2} = dx$$
$$\tan^{-1} y = x + C$$
$$y = \tan(x + C)$$
$$f(0) = 0 \Rightarrow C = 0 \ \text{[or } C = n\pi, n \in \mathbb{Z}]$$
$$f(x) = \tan x \ \text{[or } f(x) = \tan(x + n\pi)]$$

or

Method 2:

Guess that $f(x) = \tan x$

$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$
$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique $f(x) = \tan x$ is the solution.

7-7b

2. Let f be a function that is differentiable throughout its domain and that has the following properties.

(i) $f(x+y) = \frac{f(x)+f(y)}{1-f(x)f(y)}$ for all real numbers x, y, and $x+y$ in the domain of f

(ii) $\lim_{h\to 0} f(h) = 0$

(iii) $\lim_{h\to 0} \frac{f(h)}{h} = 1$

Find $f(x)$ by solving the differential equation in part (b).

Part C

1 point is earned if separates variables

1 point is earned for integration, with C

1 point is earned for c evaluated

Method 1:

$$Let\, y = f(x); \frac{dy}{dx} = 1 + y^2$$
$$\frac{dy}{1+y^2} = dx$$
$$\tan^{-1} y = x + C$$
$$y = \tan(x + C)$$
$$f(0) = 0 \Rightarrow C = 0 \;\; [\text{or } C = n\pi, n \in \mathbb{Z}]$$
$$f(x) = \tan x \;\; [or\; f(x) = \tan(x + n\pi)]$$

Method 2:

Guess that

$$f(x) = \tan x$$
$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$
$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique $f(x) = \tan x$ is the solution.

0	1	2	3

The student response earns three of the following points:

1 point is earned if separates variables

1 point is earned for integration, with C

1 point is earned for c evaluated

Method 1:

7-7b

$Let\, y = f(x); \dfrac{dy}{dx} = 1 + y^2$

$\frac{dy}{1+y^2} = dx$

$\tan^{-1} y = x + C$

$y = \tan(x + C)$

$f(0) = 0 \Rightarrow C = 0 \;\; [\text{or}\, C = n\pi, n \in \mathbb{Z}]$

$f(x) = \tan x \;[or\; f(x) = \tan(x + n\pi)]$

Method 2:

Guess that

$f\,(x) = \tan x$

$1 + [f\,(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'\,(x)$

$f\,(0) = \tan\,(0) = 0$

Since the solution to the D.E. is unique $f\,(x) = \tan x$ is the solution.

3. A particle moves along the y-axis so that its velocity v at time $t \geq 0$ is given by $v\,(t) = 1 - \tan^{-1}\left(e^t\right)$. At time $t = 0$, the particle is at $y = -1$. (Note: $\tan^{-1} x = \arctan x$)

Find the position of the particle at time $t = 2$. Is the particle moving toward the origin or away from the origin at time $t = 2$? Justify your answer.

Part D

1 point is earned for: $\int_0^2 v\,(t)dt$

1 point is earned for the handles initial condition

1 point is earned for the value of y(2)

1 point is earned for the answer with reason

$y\,(2) = \int_0^2 v\,(t)dt = -1.360 or - 1.361$

The particle is moving away from the origin since $v(2) < 0$ and $y(2) < 0$.

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for: $\int_0^2 v\,(t)dt$

1 point is earned for the handles initial condition

1 point is earned for the value of y(2)

1 point is earned for the answer with reason

$y\,(2) = \int_0^2 v\,(t)dt = -1.360 or - 1.361$

The particle is moving away from the origin since $v(2) < 0$ and $y(2) < 0$.

7-7b

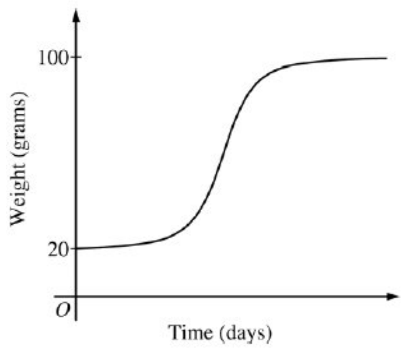
4. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

(a) Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

(b) Find $\frac{d^2B}{dt^2}$ in terms of B. Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.



(c) Use separation of variables to find $y = B(t)$, the particular sol*u*tion to the differential equation with initial condition $B(0) = 20$.

Part A

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

$$\left.\frac{dB}{dt}\right|_{B=40} = \frac{1}{5}(60) = 12$$
$$\left.\frac{dB}{dt}\right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left.\frac{dB}{dt}\right|_{B=40} > \left.\frac{dB}{dt}\right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

$$\left.\frac{dB}{dt}\right|_{B=40} = \frac{1}{5}(60) = 12$$
$$\left.\frac{dB}{dt}\right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left.\frac{dB}{dt}\right|_{B=40} > \left.\frac{dB}{dt}\right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

Part B

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

7-7b

$$\frac{d^2B}{dt^2} = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100-B) = -\frac{1}{25}(100-B)$$

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.

0	1	2 ✓

The student earns all of the following points:

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100-B) = -\frac{1}{25}(100-B)$$

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.

Part C

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

$\frac{dB}{dt} = \frac{1}{5}(100-B)$
$\int \frac{1}{100-B}dB = \int \frac{1}{5}dt$
$-\ln|100-B| = \frac{1}{5}t + c$
Because $20 \leq B < 100,\ |100-B| = 100-B$.
$-\ln(100-20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$
$100 - B = 80e^{-t/5}$
$B(t) = 100 - 80e^{-t/5},\ t \geq 0$

0	1	2	3	4	5 ✓

The student earns all of the following points:

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

7-7b

$\frac{dB}{dt} = \frac{1}{5}(100 - B)$

$\displaystyle\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$

$-\ln|100 - B| = \frac{1}{5}t + c$

Because $20 \le B < 100,\ |100 - B| = 100 - B.$

$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$

$100 - B = 80e^{-t/5}$

$B(t) = 100 - 80e^{-t/5},\ t \ge 0$

7-7b

5. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.

Part C

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$
$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$
$$-\ln\left|100 - B\right| = \frac{1}{5}t + C$$
$$\textit{Because } 20 \le B < 100,\ |100 - B| = 100 - B.$$
$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$
$$100 - B = 80e^{-t/5}$$
$$B(t) = 100 - 80e^{-t/5},\ t \ge 0$$

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

7-7b

$\frac{dB}{dt} = \frac{1}{5}(100 - B)$

$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$

$-\ln\left|100 - B\right| = \frac{1}{5}t + C$

Because $20 \le B < 100,\ |100 - B| \ = \ 100 - B.$

$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$

$100 - B = 80e^{-t/5}$

$B(t) = 100 - 80e^{-t/5},\ \ t \ge 0$

7-7b

6. A water tank at Camp Newton holds 1200 gallons of water at time $t = 0$. During the time interval $0 \le t \le 18$ hours, water is pumped into the tank at the rate

$$W(t) = 95\sqrt{t}\sin^2\left(\frac{t}{6}\right) \text{ gallons per hour.}$$

During the same time interval, water is removed from the tank at the rate

$$R(t) = 275\sin^2\left(\frac{t}{3}\right) \text{ gallons per hour.}$$

(a) Is the amount of water in the tank increasing at time $t = 15$? Why or why not?

(b) To the nearest whole number, how many gallons of water are in the tank at time $t = 18$?

(c) At what time t, $0 \le t \le 18$, is the amount of water in the tank at an absolute minimum? Show the work that leads to your conclusion.

(d) For $t > 18$, no water is pumped into the tank, but water continues to be removed at the rate $R(t)$ until the tank becomes empty. Let k be the time at which the tank becomes empty. Write, but do not solve, an equation involving an integral expression that can be used to find the value of k.

Part A

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t = 15$ since $W(15) - R(15) = -121.09 < 0$.

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t = 15$ since $W(15) - R(15) = -121.09 < 0$.

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t))dt = 1309.788$$

1310 gallons

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t))dt = 1309.788$$

1310 gallons

Part C

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t = 6.494$ or 6.495

7-7b

1 point is earned for the analysis for absolute minimum

$W(t) - R(t) = 0$

$t = 0, 6.4948, 12.9748$

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t = 6.494$ or 6.495.

0	1	2	3

The student response earns all of the following points:

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t = 6.494$ or 6.495

1 point is earned for the analysis for absolute minimum

$W(t) - R(t) = 0$

$t = 0, 6.4948, 12.9748$

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t = 6.494$ or 6.495.

Part D

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^{k} R(t)dt = 1310$$

0	1	2

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^{k} R(t)dt = 1310$$

7-7b

7. A water tank at Camp Newton holds 1200 gallons of water at time t = 0. During the time interval $0 \le t \le 18$ hours, water is pumped into the tank at the rate

$$W(t) = 95\sqrt{t}\sin^2\left(\frac{t}{6}\right) \text{ gallons per hour.}$$

During the same time interval, water is removed from the tank at the rate

$$R(t) = 275\sin^2\left(\frac{t}{3}\right) \text{ gallons per hour.}$$

To the nearest whole number, how many gallons of water are in the tank at time $t=18$?

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t))dt = 1309.788$$

1310 gallons

0	1	2	3

The student response earns three of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t))dt = 1309.788$$

1310 gallons

7-7b

Let f be the function that is defined for all real numbers x and that has the following properties.

(i) $f''(x)=24x-18$

(ii) $f'(1)=-6$

(iii) $f(2)=0$

8. Find each x such that the line tangent to the graph of f at $(x,f(x))$ is horizontal.

Part A

1 point is earned for antiderivative with or without c

1 point is earned for $f'(1) = -6$ for student's $f'(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$f'(x) = 12x^2 - 18x + C$
$f'(1) = -6 = 12 - 18 + C$
$\therefore C = 0$
$f'(x) = 6x(2x-3) = 0$
$\boxed{x = 0, \frac{3}{2}}$

0	1	2	3	4 ✓

The student response earns four of the following points:

1 point is earned for antiderivative with or without c

1 point is earned for $f'(1) = -6$ for student's $f'(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$f'(x) = 12x^2 - 18x + C$
$f'(1) = -6 = 12 - 18 + C$
$\therefore C = 0$
$f'(x) = 6x(2x-3) = 0$
$\boxed{x = 0, \frac{3}{2}}$

7-7b

9. Write an expression for $f(x)$.

Part B

1 point is earned for antiderivative with or without C

1 point is earned for equating student's f(2) and zero

1 point is earned for solving the resulting equation

$f(x) = 4x^3 - 9x^2 + C$
$f(2) = 0 = 32 - 36 + C$
$\therefore C = 4$
$\boxed{f(x) = 4x^3 - 9x^2 + 4}$

0	1	2	3 ✓

The student response earns three of the following points:

1 point is earned for antiderivative with or without C

1 point is earned for equating student's f(2) and zero

1 point is earned for solving the resulting equation

$f(x) = 4x^3 - 9x^2 + C$
$f(2) = 0 = 32 - 36 + C$
$\therefore C = 4$
$\boxed{f(x) = 4x^3 - 9x^2 + 4}$

7-7b

10. At time $t, t \geq 0$, the volume of a sphere is increasing at a rate proportional to the reciprocal of its radius. At $t = 0$, the radius of the sphere is 1 and at $t = 15$, the radius is 2. (The volume V of a sphere with a radius r is $V = 4/3\pi r^3$.)

Find the radius of the sphere as a function of t.

Part A

1 point is earned for $\frac{dV}{dt} = \frac{k}{r}$

1 point is earned for $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

1 point is earned for separates variables

1 point is earned for integration (must have C)

1 point is earned for solving for C

1 point is earned for solving for k

1 point is earned for answer

$\frac{dV}{dt} = \frac{k}{r}$
$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
$\frac{k}{r} = 4\pi r^2 \frac{dr}{dt}$
$kdt = 4\pi r^3 dr$
$kt + C = \pi r^4$
At $t = 0, r = 1$ so $C = \pi$
At $t = 15, r = 2$, so $15k + \pi = 16\pi, k = \pi$
$\pi r^4 = \pi t + \pi$
$r = \sqrt[4]{t + 1}$

0	1	2	3	4	5	6	7

The student response earns seven of the following points:

1 point is earned for $\frac{dV}{dt} = \frac{k}{r}$

1 point is earned for $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

1 point is earned for separates variables

1 point is earned for integration (must have C)

1 point is earned for solving for C

1 point is earned for solving for k

1 point is earned for answer

7-7b

$\frac{dV}{dt} = \frac{k}{r}$
$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
$\frac{k}{r} = 4\pi r^2 \frac{dr}{dt}$
$kdt = 4\pi r^3 dr$
$kt + C = \pi r^4$
At $t = 0, r = 1$ so $C = \pi$
At $t = 15, r = 2$, so $15k + \pi = 16\pi, k = \pi$
$\pi r^4 = \pi t + \pi$
$r = \sqrt[4]{t+1}$

A particle, initially at rest, moves along the x-axis so that its acceleration at any time $t \geq 0$ is given by $a(t) = 12t^2 - 4$. The position of the particle when t=1 is $x(1)=3$.

11. Find the values of t for which the particle is at rest.

Part A

2 point is earned for $v(t)$

1 point is **deducted** for leaving answer with c without solving for it

1 point is earned for setting $v(t) = 0$

1 point is earned for finding two non-negative roots

$v(t) = 4t^3 - 4t$
$v(t) = 4t^3 - 4t = 0$
$= 4t(t^2 - 1) = 0$
Therefore $t = 0, t = 1$

0	1	2	3	4

The student response earns four of the following points:

2 point is earned for $v(t)$

1 point is **deducted** for leaving answer with c without solving for it

1 point is earned for setting $v(t) = 0$

1 point is earned for finding two non-negative roots

$v(t) = 4t^3 - 4t$
$v(t) = 4t^3 - 4t = 0$
$= 4t(t^2 - 1) = 0$
Therefore $t = 0, t = 1$

7-7b

12. Write an expression for the position $x(t)$ of the particle at any time $t \geq 0$.

Part B

2 point is earned for anti-derivative of $v(t)$

1 point is **deducted** for no constant of integration

1 point is earned for evaluating C

$$x(t) = t^4 - 2t^2 + C$$
$$3 = x(1) = 1^4 - 2.1 + C$$
$$3 = C - 1$$
$$4 = C$$
$$x(t) = t^4 - 2t^2 + 4$$

0	1	2	3

The student response earns three of the following points:

2 point is earned for anti-derivative of $v(t)$

1 point is **deducted** for no constant of integration

1 point is earned for evaluating C

$$x(t) = t^4 - 2t^2 + C$$
$$3 = x(1) = 1^4 - 2.1 + C$$
$$3 = C - 1$$
$$4 = C$$
$$x(t) = t^4 - 2t^2 + 4$$

7-7b

13. A particle moves along the x-axis so that its velocity v at time t, for $0 \le t \le 5$, is given by $v(t) = \ln(t^2 - 3t + 3)$. The particle is at position $x = 8$ at time $t = 0$.

Find the position of the particle at time $t = 2$.

Part C

1 point is earned for $\int_0^2 \ln\left(u^2 - 3u + 3\right) du$

1 point is earned if handles initial condition

1 point is earned for answer

$s(t) = s(0) + \int_0^t \ln\left(u^2 - 3u + 3\right) du$

$s(2) = 8 + \int_0^2 \ln\left(u^2 - 3u + 3\right) du$

=8.368 or 8.369

0	1	2	3

The student response earns all of the following points:

1 point is earned for $\int_0^2 \ln\left(u^2 - 3u + 3\right) du$

1 point is earned if handles initial condition

1 point is earned for answer

$s(t) = s(0) + \int_0^t \ln\left(u^2 - 3u + 3\right) du$

$s(2) = 8 + \int_0^2 \ln\left(u^2 - 3u + 3\right) du$

=8.368 or 8.369

7-7b

14. An object moves along the x-axis with initial position $x(0) = 2$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \sin\left(\frac{\pi}{3}t\right)$. What is the position of the object at time $t = 4$?

Part D

1 point is earned for integral

1 point is earned for the answer

OR

1 point is earned for $x\left(t\right) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + C$

1 point is earned for the answer

$x\left(4\right) = x\left(0\right) + \int_0^4 v\left(t\right)dt = 3.432$

OR

$x\left(t\right) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$

$x\left(4\right) = 2 + \frac{9}{2\pi} = 3.432$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

OR

1 point is earned for $x\left(t\right) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + C$

1 point is earned for the answer

$x\left(4\right) = x\left(0\right) + \int_0^4 v\left(t\right)dt = 3.432$

OR

$x\left(t\right) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$

$x\left(4\right) = 2 + \frac{9}{2\pi} = 3.432$

7-7b

A particle moves along the x-axis in such a way that its acceleration at time t for $t \geq 0$ is given by $a(t)=4\cos(2t)$. At time $t = 0$, the velocity of the particle is $v(0)=1$ and its position is $x(0)=0$.

15. Write an equation for the position $x(t)$ of the particle.

Part B

1 point is earned for integrand

1 point is earned for $x\left(t\right)=-\cos 2t+t+C$ or $x\left(t\right)=-\cos 2t+t$

1 point is earned for answer

$$x\left(t\right)=\int 2\sin 2t+1dt$$
$$x\left(t\right)=-\cos 2t+t+C$$
$$x\left(0\right)=0 \Rightarrow C=1$$
$$x\left(t\right)=-\cos 2t+t+1$$

0	1	2	3 ✓

The student response earns three of the following points:

1 point is earned for integrand

1 point is earned for $x\left(t\right)=-\cos 2t+t+C$ or $x\left(t\right)=-\cos 2t+t$

1 point is earned for answer

$$x\left(t\right)=\int 2\sin 2t+1dt$$
$$x\left(t\right)=-\cos 2t+t+C$$
$$x\left(0\right)=0 \Rightarrow C=1$$
$$x\left(t\right)=-\cos 2t+t+1$$

7-7b

16. Write an equation for the velocity $v(t)$ of the particle.

Part A

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$v(t) = \int 4cos\ 2t\ dt$
$v(t) = 2sin\ 2t + C$
$v(0) = 1 \Rightarrow C = 1$
$v(t) = 2sin\ 2t + 1$

0	1	2	3

The student response earns three of the following points:

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$v(t) = \int 4cos\ 2t\ dt$
$v(t) = 2sin\ 2t + C$
$v(0) = 1 \Rightarrow C = 1$
$v(t) = 2sin\ 2t + 1$

7-7b

For $0 \le t \le 31$, the rate of change of the number of mosquitoes on Tropical Island at time *t* days is modeled by $R(t) = 5\sqrt{t}\cos\left(\frac{t}{5}\right)$ mosquitoes per day. There are 1000 mosquitoes on Tropical Island at time *t*=0.

17. According to the model, how many mosquitoes will be on the island at time t=31? Round your answer to the nearest whole number.

Part C

1 point is earned for the integral

1 point is earned for the answer

$$1000 + \int_0^{31} R(t)dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.

0	1	2

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$1000 + \int_0^{31} R(t)dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.

7-7b

18. To the nearest whole number, what is the maximum number of mosquitoes for $0 \leq t \leq 31$? Show the analysis that leads to your conclusion.

Part D

1 point is earned for the integral

1 point is earned for the answer

1 point is earned for computing interior critical points

1 point is earned for completing analysis

$R(t)=0$ when $t=0$, $t=2.5\pi$, or $t=7.5\pi$

$R(t)>0$ on $0<t<2.5\pi$

$R(t)<0$ on $2.5\pi<t<7.5\pi$

$R(t)>0$ on $7.5\pi<t<31$

The absolute maximum number of mosquitoes occurs at $t=2.5\pi$ or at $t=31$.

$$1000+\int_0^{2.5\pi} R(t)dt=1039.357$$

There are 964 mosquitoes at $t=31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

1 point is earned for computing interior critical points

1 point is earned for completing analysis

$R(t)=0$ when $t=0$, $t=2.5\pi$, or $t=7.5\pi$

$R(t)>0$ on $0<t<2.5\pi$

$R(t)<0$ on $2.5\pi<t<7.5\pi$

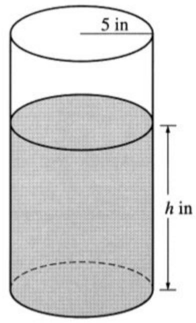
$R(t)>0$ on $7.5\pi<t<31$

The absolute maximum number of mosquitoes occurs at $t=2.5\pi$ or at $t=31$.

$$1000+\int_0^{2.5\pi} R(t)dt=1039.357,$$

There are 964 mosquitoes at $t=31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

7-7b



A coffeepot has the shape of a cylinder with radius 5 inches, as shown in the figure above. Let h be the depth of the coffee in the pot, measured in inches, where h is a function of time t, measured in seconds. The volume V of coffee in the pot is changing at the rate of $-5\pi\sqrt{h}$ cubic inches per second. (The volume V of a cylinder with radius r and height h is $V = \pi r^2 h$.)

19. Given that $h = 17$ at time $t = 0$, solve the differential equation $\frac{dh}{dt} = -\frac{\sqrt{h}}{5}$ for h as a function of t.

Part B

1 point is earned for the correct separation of variables

$\frac{\mathrm{dh}}{\mathrm{dt}} = -\frac{\sqrt{h}}{5}$

$\frac{1}{\sqrt{h}}\mathrm{dh} = -\frac{1}{5}\mathrm{dt}$

1 point is earned for the correct antiderivatives

1 point is earned for the correct constant of integration

$2\sqrt{h} = -\frac{1}{5}t + C$

1 point is earned for using initial condition $h = 17$ when $t = 0$

$2\sqrt{17} = 0 + C$

1 point is earned for correctly solving for h

$h = (-\frac{1}{10}t + \sqrt{17})^2$

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

0	1	2	3	4	5

The student response earns five of the following points:

1 point is earned for the correct separation of variables

$\frac{\mathrm{dh}}{\mathrm{dt}} = -\frac{\sqrt{h}}{5}$

$\frac{1}{\sqrt{h}}\mathrm{dh} = -\frac{1}{5}\mathrm{dt}$

1 point is earned for the correct antiderivatives

1 point is earned for the correct constant of integration

$2\sqrt{h} = -\frac{1}{5}t + C$

1 point is earned for using initial condition $h = 17$ when $t = 0$

7-7b

$2\sqrt{17} = 0 + C$

1 point is earned for correctly solving for h

$h = (-\frac{1}{10}t + \sqrt{17})^2$

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

20. At what time t is the coffeepot empty?

Part C

1 point is earned for the correct answer

$(-\frac{1}{10}t + \sqrt{17})^2 = 0$

$t = 10\sqrt{17}$

0	1

The student response earns one of the following points:

1 point is earned for the correct answer

$(-\frac{1}{10}t + \sqrt{17})^2 = 0$

$t = 10\sqrt{17}$

7-7b

21. The tide removes sand from Sandy Point Beach at a rate modeled by the function R, given by $R(t) = 2 + 5\sin\left(\frac{4\pi t}{25}\right)$.

A pumping station adds sand to the beach at a rate modeled by the function S, given by $S(t) = \frac{15t}{1+3t}$.

Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \leq t \leq 6$. At time $t = 0$, the beach contains 2500 cubic yards of sand.

Write an expression for $Y(t)$, the total number of cubic yards of sand on the beach at time t.

Part B

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x))dx$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

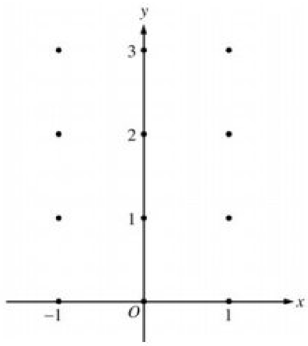
1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x))dx$$

7-7b

22. Consider the differential equation $\frac{dy}{dx} = x^4(y-2)$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



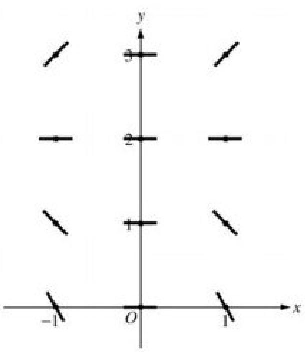
(b) While the slope field in part (a) is drawn at only twelve points, it is defined at every point in the xy-plane. Describe all points in the xy-plane for which the slopes are negative.

(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(0) = 0$.

Part A

1 point is earned for zero slope at each point (x, y) where $x = 0$ or $y = 2$

1 point is earned for positive slope at each point (x, y) where $x \neq 0$ and $y > 2$ **OR** negative slope at each point (x, y) where $x \neq 0$ and $y < 2$

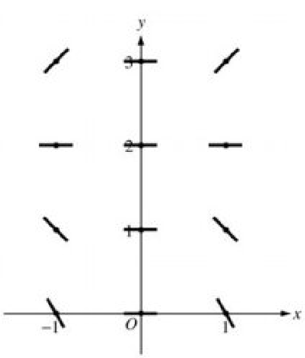


0	1	2

The student response earns all of the following points:

1 point is earned for zero slope at each point (x, y) where $x = 0$ or $y = 2$

1 point is earned for positive slope at each point (x, y) where $x \neq 0$ and $y > 2$ **OR** negative slope at each point (x, y) where $x \neq 0$ and $y < 2$



7-7b

Part B

1 point is earned for the description

Slopes are negative at points (x, y) where $x \neq 0$ and $y < 2$.

0	1

The student response earns all of the following points:

1 point is earned for the description

Slopes are negative at points (x, y) where $x \neq 0$ and $y < 2$.

Part C

1 point is earned if separates variables

2 points are earned for the antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y, 0/1 if y is not exponential

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

Solution:

$$\frac{1}{y-2}dy = x^4dx$$
$$\ln\left|y-2\right| = \frac{1}{5}x^5 + C$$
$$\left|y-2\right| = e^C e^{\frac{1}{5}x^5}$$
$$y-2 = Ke^{\frac{1}{5}x^5},\ K = \pm e^C$$
$$-2 = Ke^0 = K$$
$$y = 2 - 2e^{\frac{1}{5}x^5}$$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned if separates variables

2 points are earned for the antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned is solves for y, 0/1 if y is not exponential

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

Solution:

7-7b

$\frac{1}{y-2}dy = x^4 dx$

$\ln\left|y-2\right| = \frac{1}{5}x^5 + C$

$\left|y-2\right| = e^C e^{\frac{1}{5}x^5}$

$y - 2 = Ke^{\frac{1}{5}x^5},\ K = \pm e^C$

$-2 = Ke^0 = K$

$y = 2 - 2e^{\frac{1}{5}x^5}$

7-7b

23. Consider the differential equation $\frac{dy}{dx} = x^4(y-2)$.

Find the particular solution $y=f(x)$ to the given differential equation with the initial condition $f(0)=0$.

Part C

1 point is earned if separates variables

2 points are earned for the antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned is solves for y, 0/1 if y is not exponential

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

Solution:

$\frac{1}{y-2}dy = x^4dx$
$\ln|y-2| = \frac{1}{5}x^5 + C$
$|y-2| = e^{C}e^{\frac{1}{5}x^5}$
$y-2 = Ke^{\frac{1}{5}x^5},\ K = \pm e^{C}$
$-2 = Ke^0 = K$
$y = 2 - 2e^{\frac{1}{5}x^5}$

0	1	2	3	4	5	6

The student response earns all of the following points:

- ☐ 1 point is earned if separates variables
- ☐ 2 points are earned for the antiderivatives
- ☐ 1 point is earned for constant of integration
- ☐ 1 point is earned if uses initial condition
- ☐ 1 point is earned is solves for y, 0/1 if y is not exponential

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

Solution:

$\frac{1}{y-2}dy = x^4dx$
$\ln|y-2| = \frac{1}{5}x^5 + C$
$|y-2| = e^{C}e^{\frac{1}{5}x^5}$
$y-2 = Ke^{\frac{1}{5}x^5},\ K = \pm e^{C}$
$-2 = Ke^0 = K$
$y = 2 - 2e^{\frac{1}{5}x^5}$

7-7b

24. Consider the differential equation $\frac{dy}{dx} = \frac{3x^2}{e^{2y}}$.

(a) Find a solution $y = f(x)$ to the differential equation satisfying $f(0) = \frac{1}{2}$.

(b) Find the domain and range of the function f found in part (a).

Part A

1 point is earned for separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned for constant of integration

1 point is earned for using initial condition $f(0) = \frac{1}{2}$

1 point is earned for solving for y

$$e^{2y}dy = 3x^2dx$$
$$\frac{1}{2}e^{2y} = x^3 + C_1$$
$$e^{2y} = 2x^3 + C$$
$$y = \frac{1}{2}\ln\left(2x^3 + C\right)$$
$$\frac{1}{2} = \frac{1}{2}\ln(0 + C);\ C = e$$
$$y = \frac{1}{2}\ln\left(2x^3 + e\right)$$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned for constant of integration

1 point is earned for using initial condition $f(0) = \frac{1}{2}$

1 point is earned for solving for y

$$e^{2y}dy = 3x^2dx$$
$$\frac{1}{2}e^{2y} = x^3 + C_1$$
$$e^{2y} = 2x^3 + C$$
$$y = \frac{1}{2}\ln\left(2x^3 + C\right)$$
$$\frac{1}{2} = \frac{1}{2}\ln(0 + C);\ C = e$$
$$y = \frac{1}{2}\ln\left(2x^3 + e\right)$$

Part B

1 point is earned for $2x^3 + e > 0$

1 point is earned for domain

7-7b

Domain: $2x^3 + e > 0$

$$x^3 > -\frac{1}{2}e$$

$$x > \left(-\frac{1}{2}e\right)^{1/3} = -\left(\frac{1}{2}e\right)^{1/3}$$

1 point is earned for range

Range: $-\infty < y < \infty$

0	1	2	3

The student response earns all of the following points:

1 point is earned for $2x^3 + e > 0$

1 point is earned for domain

Domain: $2x^3 + e > 0$

$$x^3 > -\frac{1}{2}e$$

$$x > \left(-\frac{1}{2}e\right)^{1/3} = -\left(\frac{1}{2}e\right)^{1/3}$$

1 point is earned for range

Range: $-\infty < y < \infty$

7-7b

25. Consider the differential equation $\frac{dy}{dx} = \frac{3x^2}{e^{2y}}$.

Find the solution $y = f(x)$ to the differential equation satisfying $f(0) = \frac{1}{2}$.

Part A

1 point is earned for separates variables

1 point is earned for antiderivative of *dy* term

1 point is earned for antiderivative of *dx* term

1 point is earned for constant of integration

1 point is earned for uses initial condition $f(0) = \frac{1}{2}$

1 point is earned for solves for *y*

$e^{2y}dy = 3x^2dx$
$\frac{1}{2}e^{2y} = x^3 + C_1$
$e^{2y} = 2x^3 + C$
$y = \frac{1}{2}\ln\left(2x^3 + C\right)$
$\frac{1}{2} = \frac{1}{2}\ln(0 + C); C = e$
$y = \frac{1}{2}\ln\left(2x^3 + e\right)$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for separates variables

1 point is earned for antiderivative of *dy* term

1 point is earned for antiderivative of *dx* term

1 point is earned for constant of integration

1 point is earned for uses initial condtion $f(0) = \frac{1}{2}$

1 point is earned for solves for *y*

$e^{2y}dy = 3x^2dx$
$\frac{1}{2}e^{2y} = x^3 + C_1$
$e^{2y} = 2x^3 + C$
$y = \frac{1}{2}\ln\left(2x^3 + C\right)$
$\frac{1}{2} = \frac{1}{2}\ln(0 + C); C = e$
$y = \frac{1}{2}\ln\left(2x^3 + e\right)$

7-7b

26. The function f is differentiable for all real numbers. The point $\left(3, \frac{1}{4}\right)$ is on the graph of $y = f(x)$, and the slope at each point (x, y) on the graph is given by $\frac{dy}{dx} = y^2(6 - 2x)$.

(a) Find $\frac{d^2y}{dx^2}$ and evaluate it at the point $\left(3, \frac{1}{4}\right)$.

(b) Find $y = f(x)$ by solving the differential equation $\frac{dy}{dx} = y^2(6 - 2x)$ with the initial condition $f(3) = 1/4$.

Part A

2 points are earned for $\frac{d^2y}{dx^2}$

<-2> product rule or chain rule error

1 point is earned for the value at $\left(3, \frac{1}{4}\right)$

$$\frac{d^2y}{dx^2} = 2y\frac{dy}{dx}(6 - 2x) - 2y^2$$
$$= 2y^3(6 - 2x)^2 - 2y^2$$
$$\left.\frac{d^2y}{dx^2}\right|_{\left(3, \frac{1}{4}\right)} = 0 - 2\left(\frac{1}{4}\right)^2 = -\frac{1}{8}$$

0	1	2	3 ✓

The student response earns all of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

<-2> product rule or chain rule error

1 point is earned for the value at $\left(3, \frac{1}{4}\right)$

$$\frac{d^2y}{dx^2} = 2y\frac{dy}{dx}(6 - 2x) - 2y^2$$
$$= 2y^3(6 - 2x)^2 - 2y^2$$
$$\left.\frac{d^2y}{dx^2}\right|_{\left(3, \frac{1}{4}\right)} = 0 - 2\left(\frac{1}{4}\right)^2 = -\frac{1}{8}$$

Part B

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $f(3) = 1/4$

1 point is earned for solves for y

7-7b

$\frac{1}{y^2}dy = (6-2x)dx$
$-\frac{1}{y} = 6x - x^2 + C$
$-4 = 18 - 9 + C = 9 + C$
$C = -13$
$y = \frac{1}{x^2-6x+13}$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivative of *dy* term

1 point is earned for the antiderivative of *dx* term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $f(3) = 1/4$

1 point is earned for solves for *y*

$\frac{1}{y^2}dy = (6-2x)dx$
$-\frac{1}{y} = 6x - x^2 + C$
$-4 = 18 - 9 + C = 9 + C$
$C = -13$
$y = \frac{1}{x^2-6x+13}$

7-7b

27. The function f is differentiable for all real numbers. The point $\left(3, \frac{1}{4}\right)$ is on the graph of $y = f(x)$, and the slope at each point (x, y) on the graph is given by $\frac{dy}{dx} = y^2\,(6 - 2x)$.

Find $y = f(x)$ by solving the differential equation $\frac{dy}{dx} = y^2\,(6 - 2x)$ with the initial condition $f(3) = 1/4$.

Part B

1 point is earned for the separates variables

1 point is earned for the antiderivative of *dy* term

1 point is earned for the antiderivative of *dx* term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition f(3) = 1/4

1 point is earned for solves for *y*

$\frac{1}{y^2}dy = (6-2x)dx$
$= -\frac{1}{y} = 6x - x^2 + C$
$-4 = 18 - 9 + C = 9 + C$
$C = -13$
$y = \frac{1}{x^2 - 6x + 13}$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivative of *dy* term

1 point is earned for the antiderivative of *dx* term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition f(3) = 1/4

1 point is earned for solves for *y*

$\frac{1}{y^2}dy = (6-2x)dx$
$= -\frac{1}{y} = 6x - x^2 + C$
$-4 = 18 - 9 + C = 9 + C$
$C = -13$
$y = \frac{1}{x^2 - 6x + 13}$

7-7b

28. Consider the differential equation dy/dx=−xy^2/2 . Let $y = f(x)$ be the particular solution to this differential equation with the initial condition $f(-1)=2$. Find the solution $y=f(x)$ to the given differential equation with the initial condition $f(-1)=2$.

Part C

1 point is earned if separates variables

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 3/6 [1-1-1-0-0-0] if no constant of integration

Note: max 0/6 if no separation of variables

$$\frac{1}{y^2}dy = -\frac{x}{2}dx$$
$$-\frac{1}{y} = -\frac{x^2}{4} + c$$
$$-\frac{1}{2} = -\frac{1}{4} + c;\ \text{c=} -\frac{1}{4}$$
$$y = \frac{1}{\frac{x^2}{4}+\frac{1}{4}} = \frac{4}{x^2+1}$$

0	1	2	3	4	5	6

The student response earns six of the following points:

1 point is earned if separates variables

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 3/6 [1-1-1-0-0-0] if no constant of integration

Note: max 0/6 if no separation of variables

$$\frac{1}{y^2}dy = -\frac{x}{2}dx$$
$$-\frac{1}{y} = -\frac{x^2}{4} + c$$
$$-\frac{1}{2} = -\frac{1}{4} + c;\ \text{c=} -\frac{1}{4}$$
$$y = \frac{1}{\frac{x^2}{4}+\frac{1}{4}} = \frac{4}{x^2+1}$$

7-7b

29. Consider the differential equation $dy/dx = (y\text{-}1)^2\cos(\pi x)$.

Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(1) = 0$.

Part C

1 point is earned for separating the variables.

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned for answer

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note:0/6 if no separation of variables

$\frac{1}{(y-1)^2}dy = \cos(\pi x)dx$

$-(y-1)^{-1} = \frac{1}{\pi}\sin(\pi x) + C$

$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + C$

$1 = \frac{1}{\pi}\sin(\pi) + C = C$

$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + 1$

$\frac{\pi}{1-y} = \sin(\pi x) + \pi$

$y = 1 - \frac{\pi}{\sin(\pi x)+\pi}\, for\ -\infty < x < \infty$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for separating the variables.

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned for answer

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note:0/6 if no separation of variables

$\frac{1}{(y-1)^2}dy = \cos(\pi x)dx$

$-(y-1)^{-1} = \frac{1}{\pi}\sin(\pi x) + C$

$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + C$

$1 = \frac{1}{\pi}\sin(\pi) + C = C$

$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + 1$

$\frac{\pi}{1-y} = \sin(\pi x) + \pi$

$y = 1 - \frac{\pi}{\sin(\pi x)+\pi}\, for\ -\infty < x < \infty$

7-7b

30. Consider the differential equation $\frac{dy}{dx} = (3 - y)\cos x$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(0) = 1$. The function f is defined for all real numbers.

Find $y = f(x)$, the particular solution to the differential equation with the initial condition $f(0)=1$.

Part C

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

$$\frac{dy}{dx} = (3 - y)cosx$$

$$\int \frac{dy}{3 - y} = \int \cos x dx$$

$$-ln\,|3 - y| = sinx + C$$

$$-ln2 \;=\; sin\,0 + C \;\Rightarrow C \;=\; -\;ln2$$

$$-\;ln\;2\;\;-ln\;\;|3 - y|\;\;=\;\;sinx -\;ln\;2$$

Because y(0) = 1, y<3, so |3-y| = 3-y

$$3 - y = \;2e^{-sinx}$$
$$y = 3 - 2e^{-sinx}$$

Note: this solution is valid for all real numbers.

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

$$\frac{dy}{dx} = (3 - y)cosx$$

$$\int \frac{dy}{3 - y} = \int \cos x dx$$

$$-ln\,|3 - y| = sinx + C$$

$$-ln2 \;=\; sin\,0 + C \;\Rightarrow C \;=\; -\;ln2$$

$$-\;ln\;2\;\;-ln\;\;|3 - y|\;\;=\;\;sinx -\;ln\;2$$

Because y(0) = 1, y<3, so |3-y| = 3-y

7-7b

$3 - y = 2e^{-sinx}$
$y = 3 - 2e^{-sinx}$

Note: this solution is valid for all real numbers.

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

7-7b

31. Consider the differential equation $\frac{dy}{dx} = \frac{1+y}{x}$, where $x \neq 0$.

Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(-1) = 1$ and state its domain.

Part B

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for *y*

1 point is earned for the domain

$\frac{1}{1+y}dy = \frac{1}{x}dx$
$\ln|1+y| = \ln|x| + K$
$|1+y| = e^{\ln|x|+K}$
$1+y = C|x|$
$2 = C$
$1+y = 2|x|$
$y = 2|x| - 1, and, x < 0$
or
$y = -2x - 1, and, x < 0$

0	1	2	3	4	5	6	7

The student response earns all of the following points:

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for *y*

1 point is earned for the domain

$\frac{1}{1+y}dy = \frac{1}{x}dx$
$\ln|1+y| = \ln|x| + K$
$|1+y| = e^{\ln|x|+K}$
$1+y = C|x|$
$2 = C$
$1+y = 2|x|$
$y = 2|x| - 1, and, x < 0$
or
$y = -2x - 1, and, x < 0$

7-7b

32. Consider the differential equation dy/dx = $e^y(3x^2 - 6x)$. Let $y = f(x)$ be the particular solution to the differential equation that passes through (1, 0) . Find $y=f(x)$, the particular solution to the differential equation that passes through (1, 0).

Part B

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables.

$\frac{dy}{e^y} = (3x^2 - 6x)dx$

$\int \frac{dy}{e^y} = (3x^2 - 6x)dx$

$-e^{-y} = x^3 - 3x^2 + c$

$-e^{-0} = 1^3 - 3.1^2 + C \Rightarrow C = 1$

$-e^{-y} = x^3 - 3x^2 + 1$

$-e^{-y} = x^3 - 3x^2 - 1$

$-y = ln\left(-x^3 + 3x^2 - 1\right)$

$y = -ln\left(-x^3 + 3x^2 - 1\right)$

Note: This solution is valid on an interval containing $x = 1$ for which $-x^3 + 3x^2 - 1 > 0$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables.

$\frac{dy}{e^y} = (3x^2 - 6x)dx$

$\int \frac{dy}{e^y} = (3x^2 - 6x)dx$

$-e^{-y} = x^3 - 3x^2 + c$

$-e^{-0} = 1^3 - 3.1^2 + C \Rightarrow C = 1$

7-7b

$-e^{-y} = x^3 - 3x^2 + 1$

$-e^{-y} = x^3 - 3x^2 - 1$

$-y = ln\left(-x^3 + 3x^2 - 1\right)$

$y = -ln\left(-x^3 + 3x^2 - 1\right)$

Note: This solution is valid on an interval containing $x = 1$ for which $-x^3 + 3x^2 - 1 > 0$

7-7b

33. Solutions to the differential equation $dy/dx = xy^3$ also satisfy $d^2y / dx^2 = y^3(1 + 3x^2y^2)$. Let y = f(x) be a particular solution to the differential equation dy / dx = xy^3 with $f(1) = 2$.

Find the particular solution $y=f(x)$ with initial condition $f(1)=2$.

Part C

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$$\frac{dy}{dx} = xy^3$$
$$\int \frac{1}{y^3}\,dy = \int x\,dx$$
$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$
$$-\frac{1}{2.\ 2^2} = \frac{1^2}{2} + C \Rightarrow C = -\frac{5}{8}$$
$$y^2 = \frac{1}{\frac{5}{4} - x^2}$$
$$f(x) = \frac{2}{\sqrt{5-4x^2}}, \frac{-\sqrt{5}}{2} < x < \frac{\sqrt{5}}{2}$$

0	1	2	3	4	5

The student response earns none of the following points:

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

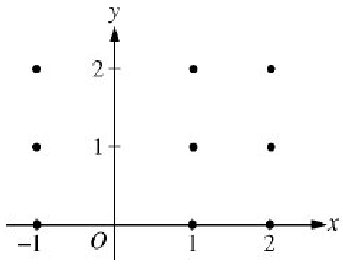
Note: 0/5 if no separation of variables

$$\frac{dy}{dx} = xy^3$$
$$\int \frac{1}{y^3}\,dy = \int x\,dx$$
$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$
$$-\frac{1}{2.\ 2^2} = \frac{1^2}{2} + C \Rightarrow C = -\frac{5}{8}$$
$$y^2 = \frac{1}{\frac{5}{4} - x^2}$$
$$f(x) = \frac{2}{\sqrt{5-4x^2}}, \frac{-\sqrt{5}}{2} < x < \frac{\sqrt{5}}{2}$$

7-7b

34. Consider the differential equation $\frac{dy}{dx} = \frac{y-1}{x^2}$, where $x \neq 0$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



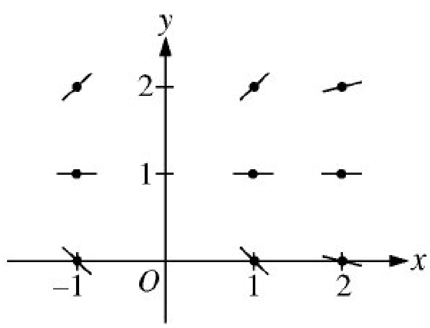
(b) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(2) = 0$.

(c) For the particular solution $y = f(x)$ described in part (b), find $\lim_{x \to \infty} f(x)$.

Part A

1 point: For zero slopes

1 point: For all other slopes

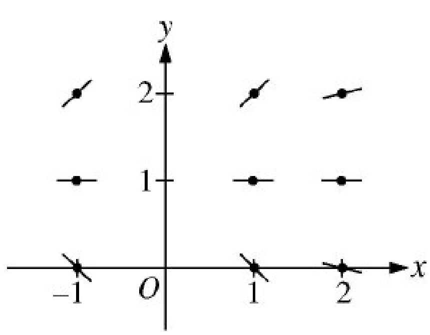


0	1	2 ✓

The response earns all of the following points:

1 point: For zero slopes

1 point: For all other slopes



7-7b

Part B

The response can earn up to 6 points:

Note: 0/6 if no separation of variables

Note: max 3/6 if no constant of integration

1 point: For separation of variables

2 points: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

1 point: For the solution of y

$\frac{1}{y-1}dy = \frac{1}{x^2}dx$
$\ln|y-1| = -\frac{1}{x} + C$
$|y-1| = e^{-\frac{1}{x}+C}$
$|y-1| = e^{C}e^{-\frac{1}{x}}$
$y-1 = ke^{-\frac{1}{x}}$, where $k = \pm e^{C}$
$-1 = ke^{-\frac{1}{2}}$
$k = -e^{\frac{1}{2}}$
$f(x) = 1 - e^{\left(\frac{1}{2}-\frac{1}{x}\right)},\ x > 0$

0	1	2	3	4	5	6

The response earns all of the following points:

Note: 0/6 if no separation of variables

Note: max 3/6 if no constant of integration

1 point: For separation of variables

2 points: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

1 point: For the solution of y

$\frac{1}{y-1}dy = \frac{1}{x^2}dx$
$\ln|y-1| = -\frac{1}{x} + C$
$|y-1| = e^{-\frac{1}{x}+C}$
$|y-1| = e^{C}e^{-\frac{1}{x}}$
$y-1 = ke^{-\frac{1}{x}}$, where $k = \pm e^{C}$
$-1 = ke^{-\frac{1}{2}}$
$k = -e^{\frac{1}{2}}$
$f(x) = 1 - e^{\left(\frac{1}{2}-\frac{1}{x}\right)},\ x > 0$

7-7b

Part C

The response can earn up to 1 point:

1 point: For the correct limit

$\lim_{x\to\infty} 1 - e^{\left(\frac{1}{2}-\frac{1}{x}\right)} = 1 - \sqrt{e}$

0	1 ✓

The response earns the following point:

1 point: For the correct limit

$\lim_{x\to\infty} 1 - e^{\left(\frac{1}{2}-\frac{1}{x}\right)} = 1 - \sqrt{e}$

35. Consider the differential equation $\frac{dy}{dx} = \frac{y-1}{x^2}$, where $x \neq 0$.

Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(2) = 0$.

FRQ Score

The response can earn up to 6 points:

Note: 0/6 if no separation of variables

Note: max 3/6 if no constant of integration.

1 point: For separation of variables

2 point: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

1 point: For the solution of y

$\frac{1}{y-1}dy = \frac{1}{x^2}dx \ln|y-1| = -\frac{1}{x} + C\, |y-1| = e^{-\frac{1}{x}+C}\, |y-1| = e^C e^{-\frac{1}{x}} y - 1 = ke^{-\frac{1}{x}}, \text{where } k\ = \pm e^C - 1 = ke^{-\frac{1}{2}}k\ = -e^{\frac{1}{2}} f(x) = 1 - e^{\left(\frac{1}{2}-\frac{1}{x}\right)}, x > 0$

0	1	2	3	4	5	6 ✓

The response earns all six of the following points:

1 point: For separation of variables

2 point: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

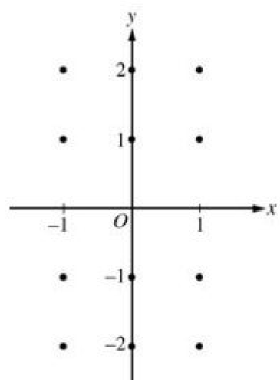
1 point: For the solution of y

$\frac{1}{y-1}dy = \frac{1}{x^2}dx \ln|y-1| = -\frac{1}{x} + C\, |y-1| = e^{-\frac{1}{x}+C}\, |y-1| = e^C e^{-\frac{1}{x}} y - 1 = ke^{-\frac{1}{x}}, \text{where } k\ = \pm e^C - 1 = ke^{-\frac{1}{2}}k\ = -e^{\frac{1}{2}} f(x) = 1 - e^{\left(\frac{1}{2}-\frac{1}{x}\right)}, x > 0$

7-7b

36. Consider the differential equation $\frac{dy}{dx} = -\frac{2x}{y}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



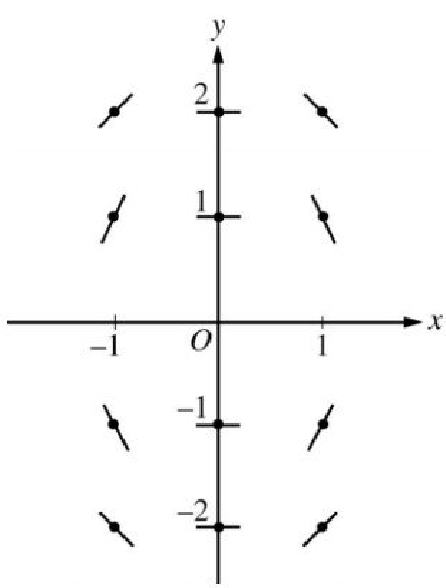
(b) Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(1) = -1$. Write an equation for the line tangent to the graph of f at $(1, -1)$ and use it to approximate $f(1.1)$.

(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(1) = -1$.

Part A

1 point is earned for the zero slopes

1 point is earned for the nonzero slopes



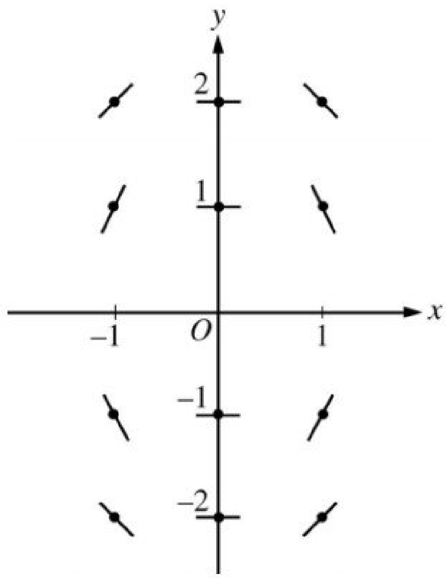
0	1	2

The student response earns all of the following points:

1 point is earned for the zero slopes

1 point is earned for the nonzero slopes

7-7b



Part B

1 point is earned for the equation of the tangent line

1 point is earned for the approximation for $f(1.1)$

The line tangent to f at (1, -1) is $y + 1 = 2(x - 1)$. Thus, $f(1.1)$ is approximately -0.8.

0	1	2

The student response earns all of the following points:

1 point is earned for the equation of the tangent line

1 point is earned for the approximation for $f(1.1)$

The line tangent to f at (1, -1) is $y + 1 = 2(x - 1)$. Thus, $f(1.1)$ is approximately -0.8.

Part C

1 point is earned for the separates variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for y

$$\frac{dy}{dx} = -\frac{2x}{y}$$
$$ydy = -2xdx$$
$$\frac{y^2}{2} = -x^2 + C$$
$$\frac{1}{2} = -1 + C;\ C = \frac{3}{2}$$
$$y^2 = -2x^2 + 3$$

Since the particular solution goes through (1, -1), y must be negative. Thus the particular solution is $y = -\sqrt{3 - 2x^2}$.

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0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for *y*

$$\frac{dy}{dx} = -\frac{2x}{y}$$
$$ydy = -2xdx$$
$$\frac{y^2}{2} = -x^2 + C$$
$$\frac{1}{2} = -1 + C;\ C = \frac{3}{2}$$
$$y^2 = -2x^2 + 3$$

Since the particular solution goes through (1, -1), *y* must be negative. Thus the particular solution is $y = -\sqrt{3 - 2x^2}$.

7-7b

37. Solutions to the differential equation $\frac{dy}{dx} = xy^3$ also satisfy $\frac{d^2y}{dx^2} = y^3\left(1+3x^2y^2\right)$. Let y = f(x) be a particular solution to the differential equation $\frac{dy}{dx} = xy^3$ with f(1) = 2.

(a) Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 1$.

(b) Use the tangent line equation from part (a) to approximate $f(1.1)$. Given that $f(x) > 0$ for $1 < x < 1.1$, is the approximation for $f(1.1)$ greater than or less than $f(1.1)$? Explain your reasoning.

(c) Find the particular solution $y = f(x)$ with initial condition $f(1) = 2$.

Part A

1 point is earned for $f'(1)$

1 point is earned for the answer

$$f'(1) = \left.\frac{dy}{dx}\right|_{(1,2)} = 8$$

An equation of the tangent line is $y = 2 + 8(x - 1)$.

0	1	2

The student response earns all of the following points:

1 point is earned for $f'(1)$

1 point is earned for the answer

$$f'(1) = \left.\frac{dy}{dx}\right|_{(1,2)} = 8$$

An equation of the tangent line is $y = 2 + 8(x - 1)$.

Part B

1 point is earned for approximation

1 point is earned for conclusion with explanation

$f(1.1) \approx 2.8$

Since $y = f(x) > 0$ on the interval $1 \le x < 1.1$,

$$\frac{d^2y}{dx^2} = y^3\left(1+3x^2y^2\right) > 0 \text{ on this interval.}$$

Therefore on the interval $1 < x < 1.1$, the line tangent to the graph of $y = f(x)$ at $x = 1$ lies below the curve and the approximation 2.8 is less than $f(1.1)$

0	1	2

The student response earns all of the following points:

1 point is earned for approximation

1 point is earned for conclusion with explanation

$f(1.1) \approx 2.8$

Since $y = f(x) > 0$ on the interval $1 \le x < 1.1$,

7-7b

$\frac{d^2y}{dx^2} = y^3\left(1+3x^2y^2\right) > 0$ on this interval.

Therefore on the interval $1 < x < 1.1$, the line tangent to the graph of $y = f(x)$ at $x = 1$ lies below the curve and the approximation 2.8 is less than $f(1.1)$

Part C

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$$\frac{dy}{dx} = xy^3$$
$$\int \frac{1}{y^3}\,dy = \int x\,dx$$
$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$
$$-\frac{1}{2.\,2^2} = \frac{1^2}{2} + C \Rightarrow C = -\frac{5}{8}$$
$$y^2 = \frac{1}{\frac{5}{4} - x^2}$$
$$f(x) = \frac{2}{\sqrt{5-4x^2}}, \frac{-\sqrt{5}}{2} < x < \frac{\sqrt{5}}{2}$$

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

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$$\frac{dy}{dx} = xy^3$$
$$\int \frac{1}{y^3}\,dy = \int x\,dx$$
$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$
$$-\frac{1}{2.\,2^2} = \frac{1^2}{2} + C \Rightarrow C = -\frac{5}{8}$$
$$y^2 = \frac{1}{\frac{5}{4} - x^2}$$
$$f(x) = \frac{2}{\sqrt{5-4x^2}}, \frac{-\sqrt{5}}{2} < x < \frac{\sqrt{5}}{2}$$

7-7b

38. Let f be a function with $f(1) = 4$ such that for all points (x, y) on the graph of f the slope is given by $3x^2+1\ /\ 2y$.

(a) Find the slope of the graph of f at the point where $x = 1$.

(b) Write an equation for the line tangent to the graph of f at $x = 1$ and use it to approximate $f(1.2)$.

(c) Find $f(x)$ by solving the separable differential equation $dy/dx = 3x^2+1\ /\ 2y$ with the initial condition $f(1) = 4$.

(d) Use your solution from part (c) to find $f(1.2)$.

Part A

1 point is earned for the answer

$$\frac{dy}{dx} = \frac{3x^2 + 1}{2y}$$

$$\left.\frac{dy}{dx}\right|_{\substack{x=1 \\ y=4}} = \frac{3 + 1}{2 \cdot 4} = \frac{4}{8} = \frac{1}{2}$$

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{dy}{dx} = \frac{3x^2 + 1}{2y}$$

$$\left.\frac{dy}{dx}\right|_{\substack{x=1 \\ y=4}} = \frac{3 + 1}{2 \cdot 4} = \frac{4}{8} = \frac{1}{2}$$

Part B

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$y - 4 = \frac{1}{2}(x - 1)$

$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$

$f(1.2) \approx 0.1 + 4 = 4.1$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$y - 4 = \frac{1}{2}(x - 1)$

$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$

$f(1.2) \approx 0.1 + 4 = 4.1$

7-7b

Part C

1 point is earned if separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned if uses $y = 4$ when $x = 1$ to pick one function out of a family of functions

1 point is earned if solves for y

$2y\,dy = \left(3x^2 + 1\right)dx$

$\int 2y\ dy\ = \int (3x^2\ +\ 1)\ dx$

$y^2\ =\ x^3\ +\ x\ +\ C$

$4^2\ =\ 1\ +\ 1\ +\ C$

$14 = C$

$y^2\ =\ x^3\ +\ x\ +\ 14$

$y = \sqrt{x^{\,3} + x + 14}$ is branch with point (1, 4)

$f(x) = \sqrt{x^3\ + x + 14}$

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned if separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned if uses $y = 4$ when $x = 1$ to pick one function out of a family of functions

1 point is earned if solves for y

$2y\,dy = \left(3x^2 + 1\right)dx$

$\int 2y\ dy\ = \int (3x^2\ +\ 1)\ dx$

$y^2\ =\ x^3\ +\ x\ +\ C$

$4^2\ =\ 1\ +\ 1\ +\ C$

$14 = C$

7-7b

$y^2 = x^3 + x + 14$

$y = \sqrt{x^3 + x + 14}$ is branch with point (1, 4)

$f(x) = \sqrt{x^3 + x + 14}$

Part D

1 point is earned if answer, from student's solution to the given differential equation in (c)

$f(1.2) = \sqrt{1.2^3 + 1.2 + 14} \approx 4.114$

0	1

The student response earns all of the following points:

1 point is earned if answer, from student's solution to the given differential equation in (c)

$f(1.2) = \sqrt{1.2^3 + 1.2 + 14} \approx 4.114$

7-7b

39. Let f be a function with $f(1) = 4$ such that for all points (x, y) on the graph of f the slope is given by $3x^2+1$ / $2y$.
Find $f(x)$ by solving the separable differential equation $dy/dx = 3x^2+1$ / $2y$ with the initial condition $f(1) = 4$.

Part C

1 point is earned if separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned if uses $y = 4$ when $x = 1$ to pick one function out of a family of functions

1 point is earned if solves for y

$2ydy = \left(3x^2 + 1\right)dx$

$\int 2\mathrm{y}\ \mathrm{dy} = \int (3\mathrm{x}^2 + 1)\ \mathrm{dx}$

$\mathrm{y}^2 = \mathrm{x}^3 + \mathrm{x} + \mathrm{C}$

$4^2 = 1 + 1 + \mathrm{C}$

$14 = C$

$\mathrm{y}^2 = \mathrm{x}^3 + \mathrm{x} + 14$

$y = \sqrt{x^{\,3} + x + 14}$ is branch with point (1, 4)

$f(x) = \sqrt{x^3 + x + 14}$

0	1	2	3	4	5 ✓

The student response earns five of the following points:

1 point is earned if separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned if uses $y = 4$ when $x = 1$ to pick one function out of a family of functions

1 point is earned if solves for y

$2ydy = \left(3x^2 + 1\right)dx$

$\int 2\mathrm{y}\ \mathrm{dy} = \int (3\mathrm{x}^2 + 1)\ \mathrm{dx}$

$\mathrm{y}^2 = \mathrm{x}^3 + \mathrm{x} + \mathrm{C}$

$4^2 = 1 + 1 + \mathrm{C}$

$14 = C$

$\mathrm{y}^2 = \mathrm{x}^3 + \mathrm{x} + 14$

$y = \sqrt{x^{\,3} + x + 14}$ is branch with point (1, 4)

$f(x) = \sqrt{x^3 + x + 14}$

7-7b

40. Consider the differential equation $\frac{dy}{dx} = -\frac{2x}{y}$.

Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(1) = -1$.

Part C

1 point is earned for the separates variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for y

$$\frac{dy}{dx} = -\frac{2x}{y}$$
$$ydy = -2xdx$$
$$\frac{y^2}{2} = -x^2 + C$$
$$\frac{1}{2} = -1 + C; C = \frac{3}{2}$$
$$y^2 = -2x^2 + 3$$

Since the particular solution goes through (1, -1), y must be negative. Thus the particular solution is $y = -\sqrt{3 - 2x^2}$

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for y

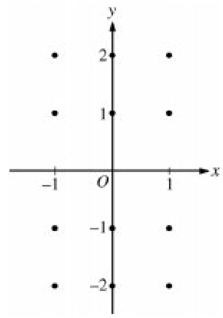
$$\frac{dy}{dx} = -\frac{2x}{y}$$
$$ydy = -2xdx$$
$$\frac{y^2}{2} = -x^2 + C$$
$$\frac{1}{2} = -1 + C; C = \frac{3}{2}$$
$$y^2 = -2x^2 + 3$$

Since the particular solution goes through (1, -1), y must be negative. Thus the particular solution is $y = -\sqrt{3 - 2x^2}$

7-7b

41. Consider the differential equation $\frac{dy}{dx} = \frac{x+1}{y}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated, and for $-1 < x < 1$, sketch the solution curve that passes through the point $(0, -1)$.



(b) While the slope field in part (a) is drawn at only twelve points, it is defined at every point in the xy-plane for which $y \neq 0$. Describe all points in the xy-plane, $y \neq 0$, for which $dy/dx = -1$.

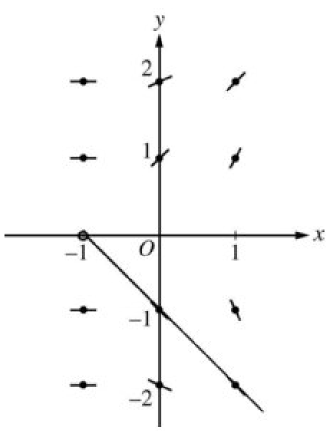
(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(0) = -2$.

Part A

1 point is earned for zero slopes

1 point is earned for nonzero slopes

1 point is earned for solution curve through (0, -1)



0	1	2	3

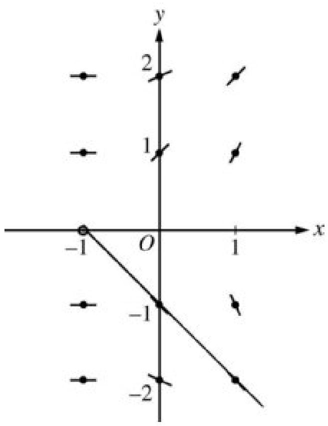
The student response earns all of the following points:

1 point is earned for zero slopes

1 point is earned for nonzero slopes

1 point is earned for solution curve through (0, -1)

7-7b



Part B

1 point is earned description

$-1 = \frac{x+1}{y} \Rightarrow y = -x - 1$
$\frac{dy}{dx} = -1$ for all (x, y) with $y = -x - 1$ and $y \neq 0$

0	1

The student response earns all of the following points:

1 point is earned description

$-1 = \frac{x+1}{y} \Rightarrow y = -x - 1$
$\frac{dy}{dx} = -1$ for all (x, y) with $y = -x - 1$ and $y \neq 0$

Part C

1 point is earned if separates variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$\int y\,dy = \int (x+1)dx$
$\frac{y^2}{2} = \frac{x^2}{2} + x + c$
$\frac{(-2)^2}{2} = \frac{0^2}{2} + 0 + c \Rightarrow c = 2$
$y^2 = x^2 + 2x + 4$

7-7b

Since the solution goes through (0, -2), y must be negative. Therefore $y = -\sqrt{x^2 + 2x + 4}$.

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned if separates variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$$\int y\,dy = \int (x+1)dx$$
$$\frac{y^2}{2} = \frac{x^2}{2} + x + c$$
$$\frac{(-2)^2}{2} = \frac{0^2}{2} + 0 + c \Rightarrow c = 2$$
$$y^2 = x^2 + 2x + 4$$

Since the solution goes through (0, -2), y must be negative. Therefore $y = -\sqrt{x^2 + 2x + 4}$.

7-7b

42. Consider the differential equation dy/dx = x+1/y.
Find the particular solution y=f(x) to the given differential equation with the initial condition f(0)=-2.

Part C

1 point is earned if separates variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$$\int y dy = \int (x+1) dx$$
$$\frac{y^2}{2} = \frac{x^2}{2} + x + c$$
$$\frac{(-2)^2}{2} = \frac{0^2}{2} + 0 + c \Rightarrow c = 2$$
$$y^2 = x^2 + 2x + 4$$

Since the solution goes through (0,-2) , y must be negative. Therefore $y = -\sqrt{x^2+2x+4}$.

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned if separates variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$$\int y dy = \int (x+1) dx$$
$$\frac{y^2}{2} = \frac{x^2}{2} + x + c$$
$$\frac{(-2)^2}{2} = \frac{0^2}{2} + 0 + c \Rightarrow c = 2$$
$$y^2 = x^2 + 2x + 4$$

Since the solution goes through (0,-2) , y must be negative. Therefore $y = -\sqrt{x^2+2x+4}$

7-7b

43. Consider the differential equation $dy/dx = e^y(3x^2 - 6x)$. Let $y = f(x)$ be the particular solution to the differential equation that passes through (1, 0).

(a) Write an equation for the line tangent to the graph of f at the point (1, 0). Use the tangent line to approximate $f(1.2)$.

(b) Find $y = f(x)$, the particular solution to the differential equation that passes through (1, 0).

Part A

1 point is earned for $\frac{dy}{dx}$ at the point $(x, y) = (1, 0)$

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left.\frac{dy}{dx}\right|_{(x,y)=(1,0)} = e^0\left(3 \cdot 1^2 - 6 \cdot 1\right) = -3$$

An equation for the tangent line is $y = -3(x - 1)$.

$f(1.2) \approx -3(1.2 - 1) = -0.6$

0	1	2	3

The student response earns all of the following points:

1 point is earned for $\frac{dy}{dx}$ at the point $(x, y) = (1, 0)$

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left.\frac{dy}{dx}\right|_{(x,y)=(1,0)} = e^0\left(3 \cdot 1^2 - 6 \cdot 1\right) = -3$$

An equation for the tangent line is $y = -3(x - 1)$.

$f(1.2) \approx -3(1.2 - 1) = -0.6$

Part B

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables.

7-7b

$$\frac{dy}{e^y} = (3x^2 - 6x)dx$$
$$\int \frac{dy}{e^y} = \int (3x^2 - 6x)dx$$
$$-e^{-y} = x^3 - 3x^2 + C$$
$$-e^{-0} = 1^3 - 3 \cdot 1^2 + C \Rightarrow C = 1$$
$$-e^{-y} = x^3 - 3x^2 + 1$$
$$e^{-y} = -x^3 + 3x^2 - 1$$
$$-y = \ln\left(-x^3 + 3x^2 - 1\right)$$
$$y = -\ln\left(-x^3 + 3x^2 - 1\right)$$

Note: This solution is valid on an interval containing $x = 1$ for which $-x^3 + 3x^2 - 1 > 0$.

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables.

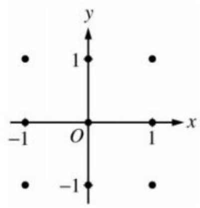
$$\frac{dy}{e^y} = (3x^2 - 6x)dx$$
$$\int \frac{dy}{e^y} = \int (3x^2 - 6x)dx$$
$$-e^{-y} = x^3 - 3x^2 + C$$
$$-e^{-0} = 1^3 - 3 \cdot 1^2 + C \Rightarrow C = 1$$
$$-e^{-y} = x^3 - 3x^2 + 1$$
$$e^{-y} = -x^3 + 3x^2 - 1$$
$$-y = \ln\left(-x^3 + 3x^2 - 1\right)$$
$$y = -\ln\left(-x^3 + 3x^2 - 1\right)$$

Note: This solution is valid on an interval containing $x = 1$ for which $-x^3 + 3x^2 - 1 > 0$.

7-7b

44. Consider the differential equation $dy/dx = (y - 1)^2\cos(\pi x)$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



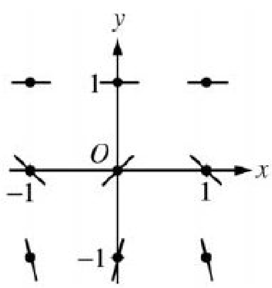
(b) There is a horizontal line with equation $y = c$ that satisfies this differential equation. Find the value of c.

(c) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(1) = 0$.

Part A

1 point is earned for the zero slopes

1 point is earned for all other slopes

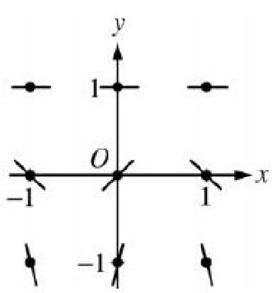


0	1	2

The student response earns all of the following points:

1 point is earned for the zero slopes

1 point is earned for all other slopes



Part B

1 point is earned for $c = 1$

The line $y = 1$ satisfies the differential equation, so $c = 1$.

0	1

The student response earns all of the following points:

1 point is earned for $c = 1$

7-7b

The line $y = 1$ satisfies the differential equation, so $c = 1$.

Part C

1 point is earned for separating the variables.

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned for answer

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note:0/6 if no separation of variables

$\frac{1}{(y-1)^2}dy = \cos(\pi x)dx$

$-(y-1)^{-1} = \frac{1}{\pi}\sin(\pi x) + C$

$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + C$

$1 = \frac{1}{\pi}\sin(\pi) + C = C$

$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + 1$

$\frac{\pi}{1-y} = \sin(\pi x) + \pi$

$y = 1 - \frac{\pi}{\sin(\pi x)+\pi}$ for $-\infty < x < \infty$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for separating the variables.

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned for answer

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note:0/6 if no separation of variables

$\frac{1}{(y-1)^2}dy = \cos(\pi x)dx$

$-(y-1)^{-1} = \frac{1}{\pi}\sin(\pi x) + C$

$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + C$

$1 = \frac{1}{\pi}\sin(\pi) + C = C$

$\frac{1}{1-y} = \frac{1}{\pi}\sin(\pi x) + 1$

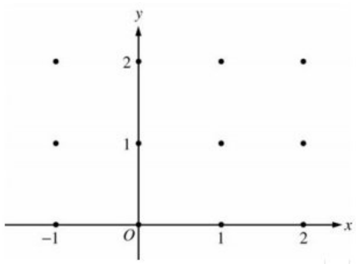
$\frac{\pi}{1-y} = \sin(\pi x) + \pi$

$y = 1 - \frac{\pi}{\sin(\pi x)+\pi}$ for $-\infty < x < \infty$

7-7b

45. Consider the differential equation $dy/dx = -xy^2/2$. Let $y = f(x)$ be the particular solution to this differential equation with the initial condition $f(-1) = 2$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



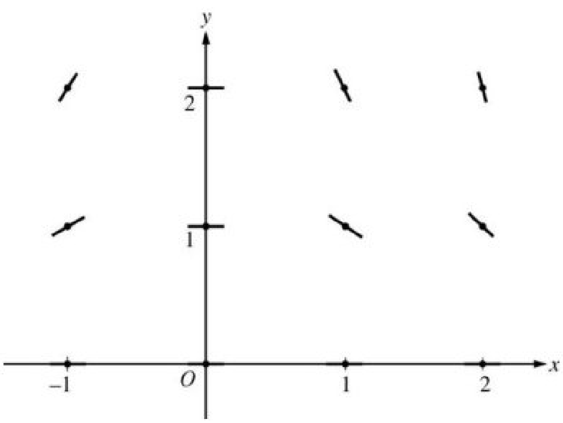
(b) Write an equation for the line tangent to the graph of f at $x = -1$.

(c) Find the solution $y = f(x)$ to the given differential equation with the initial condition $f(-1) = 2$.

Part A

1 point is earned for zero slopes

1 point is earned for nonzero slopes

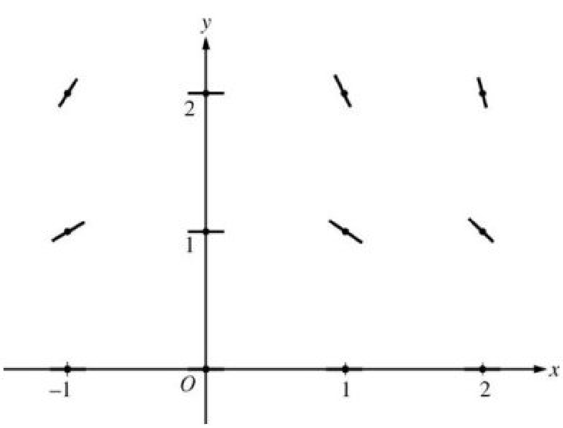


0	1	2

The student response earns all of the following points:

1 point is earned for zero slopes

1 point is earned for nonzero slopes



7-7b

Part B

1 point is earned for the equation

$$\text{Slope} = \frac{-(-1)4}{2} = 2$$
$$y - 2 = 2(x+1)$$

0	1

The student response earns all of the following points:

1 point is earned for the equation

$$\text{Slope} = \frac{-(-1)4}{2} = 2$$
$$y - 2 = 2(x+1)$$

Part C

1 point is earned if separates variables

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for *y*

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: max 0/6 if no separation of variables

$$\frac{1}{y^2}dy = -\frac{x}{2}dx$$
$$-\frac{1}{y} = -\frac{x^2}{4} + C$$
$$-\frac{1}{2} = -\frac{1}{4} + C;\ C = -\frac{1}{4}$$
$$y = \frac{1}{\frac{x^2}{4}+\frac{1}{4}} = \frac{4}{x^2+1}$$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned if separates variables

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for *y*

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: max 0/6 if no separation of variables

7-7b

$\frac{1}{y^2}dy = -\frac{x}{2}dx$

$-\frac{1}{y} = -\frac{x^2}{4} + C$

$-\frac{1}{2} = -\frac{1}{4} + C;\ C = -\frac{1}{4}$

$y = \frac{1}{\frac{x^2}{4}+\frac{1}{4}} = \frac{4}{x^2+1}$

7-7b

46. Let f and g be functions that are differentiable for all real numbers x and that have the following properties.

i. $f'(x) = f(x) - g(x)$

ii. $g'(x) = g(x) - f(x)$

iii. $f(0) = 5$

iv. $g(0) = 1$

Find $f(x)$ and $g(x)$. Show your work.

Part B

1 point is earned for the correct differential equation for one function

1 point is earned for correctly separating variables

1 point is earned for the correct antidifferentiation

1 point is earned for correctly evaluating constant

1 point is earned for correctly finding g

1 point is earned for correctly finding f

Note: Max 1/6 if not working with DE for one function

$f(x) = 6 - g(x)$ so
$g'(x) = g(x) - 6 + g(x) = 2g(x) - 6$
$\frac{dy}{dx} = 2y - 6; \frac{dy}{2y-6} = dx$
$\frac{1}{2}\ln|2y - 6| = x + C$
$\ln|2y - 6| = 2x + K$
$|2y - 6| = e^{2x+K}$
$2y - 6 = Ae^{2x}$
$x = 0 = y = 1$ so $-4 = A$
$2y = -4e^{2x} + 6$
$y = 3 - 2e^{2x} = g(x)$
$f(x) = 6 - g(x) = 3 + 2e^{2x}$

OR

1 point is earned for the correct equation in $f - g$

1 point is earned for the correct solution with constant

1 point is earned for correctly evaluating constant

1 point is earned for correctly using $f(x) + g(x) = 6$

1 point is earned for correctly finding f

1 point is earned for correctly finding g

7-7b

$f(x) = 6 - g(x)$ so
$g\prime(x) = g(x) - 6 + g(x) = 2g(x) - 6$
$g\prime(x) - 2g(x) = -6$
$\frac{d}{dx}[g(x)e^{-2x}] = -6e^{-2x}$
$g(x)e^{-2x} = \int -6\mathrm{e}^{-2x}dx$
$= 3e^{-2x} + C$
$g(x) = 3 + Ce^{2x}$
$1 = g(0) = 3 + C;\ C = -2$
$g(x) = 3 - 2e^{2x}$
$f(x) = 6 - g(x) = 3 + 2e^{2x}$

OR

Student attempts solution with infinite series

1 point is earned for the correct constant terms in f and g

1 point is earned for the correct linear terms in f and g

1 point is earned for the correct quadratic term in f

1 point is earned for the correct quadratic term in g

1 point is earned for the correct general term for f or g

1 point is earned for the correct series sum to 6

$f' - g' = 2f - 2g = 2(f - g),$ so
$f - g = Ae^{2x}$
$f(0) - g(0) = 4 = A$
$f(x) - g(x) = 4e^{2x}$
$f(x) + g(x) = 6$
$2\,f(x) = 6 + 4e^{2x}$
$f(x) = 3 + 2e^{2x}$
$g(x) = 6 - f(x) = 3 - 2e^{2x}$

0	1	2	3	4	5	6

The student response earns six of the following points:

1 point is earned for the correct differential equation for one function

1 point is earned for correctly separating variables

1 point is earned for the correct antidifferentiation

1 point is earned for correctly evaluating constant

1 point is earned for correctly finding g

1 point is earned for correctly finding f

Note: Max 1/6 if not working with DE for one function

7-7b

$f(x) = 6 - g(x)$ so
$g'(x) = g(x) - 6 + g(x) = 2g(x) - 6$
$\frac{dy}{dx} = 2y - 6; \frac{dy}{2y-6} = dx$
$\frac{1}{2}\ln\left|2y - 6\right| = x + C$
$\ln|2y - 6| = 2x + K$
$\left|2y - 6\right| = e^{2x+K}$
$2y - 6 = Ae^{2x}$
$x = 0 = y = 1$ so $-4 = A$
$2y = -4e^{2x} + 6$
$y = 3 - 2e^{2x} = g(x)$
$f(x) = 6 - g(x) = 3 + 2e^{2x}$

OR

1 point is earned for the correct equation in *f* - *g*

1 point is earned for the correct solution with constant

1 point is earned for correctly evaluating constant

1 point is earned for correctly using $f(x) + g(x) = 6$

1 point is earned for correctly finding *f*

1 point is earned for correctly finding *g*

$f(x) = 6 - g(x)$ so
$g\prime(x) = g(x) - 6 + g(x) = 2g(x) - 6$
$g\prime(x) - 2g(x) = -6$
$\frac{d}{dx}[g(x)e^{-2x}] = -6e^{-2x}$
$g(x)e^{-2x} = \int -6\mathrm{e}^{-2x}dx$
$= 3e^{-2x} + C$
$g(x) = 3 + Ce^{2x}$
$1 = g(0) = 3 + C;\ C = -2$
$g(x) = 3 - 2e^{2x}$
$f(x) = 6 - g(x) = 3 + 2e^{2x}$

OR

Student attempts solution with infinite series

1 point is earned for the correct constant terms in *f* and *g*

1 point is earned for the correct linear terms in *f* and *g*

1 point is earned for the correct quadratic term in *f*

1 point is earned for the correct quadratic term in *g*

1 point is earned for the correct general term for *f* or *g*

1 point is earned for the correct series sum to 6

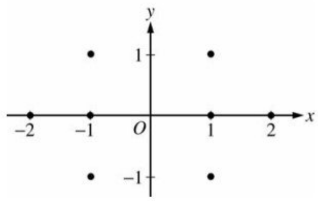
$f' - g' = 2f - 2g = 2(f - g)$, so
$f - g = Ae^{2x}$
$f(0) - g(0) = 4 = A$
$f(x) - g(x) = 4e^{2x}$
$f(x) + g(x) = 6$
$2\,f(x) = 6 + 4e^{2x}$
$f(x) = 3 + 2e^{2x}$
$g(x) = 6 - f(x) = 3 - 2e^{2x}$

7-7b

7-7b

47. Consider the differential equation $\frac{dy}{dx} = \frac{1+y}{x}$, where $x \neq 0$.

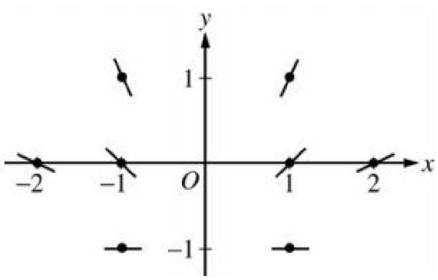
(a) On the axes provided, sketch a slope field for the given differential equation at the eight points indicated.



(b) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(-1) = 1$ and state its domain.

Part A

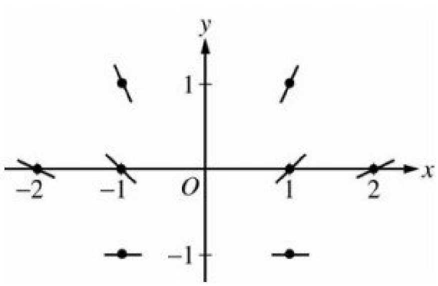
2 points are earned for the sign of slope at each point and relative steepness of slope lines in rows and columns



0	1	2

The student response earns all of the following points:

2 points are earned for the sign of slope at each point and relative steepness of slope lines in rows and columns



Part B

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

7-7b

1 point is earned for the domain

$\frac{1}{1+y}dy = \frac{1}{x}dx$
$\ln|1+y| = \ln|x| + K$
$|1+y| = e^{\ln|x|+K}$
$1+y = C|x|$
$2 = C$
$1+y = 2|x|$
$y = 2|x| - 1$ and $x < 0$
or
$y = -2x - 1$ and $x < 0$

0	1	2	3	4	5	6	7

The student response earns all of the following points:

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for *y*

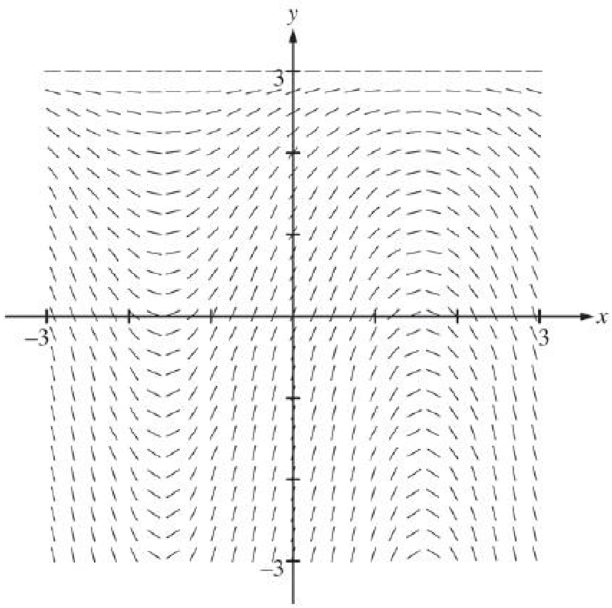
1 point is earned for the domain

$\frac{1}{1+y}dy = \frac{1}{x}dx$
$\ln|1+y| = \ln|x| + K$
$|1+y| = e^{\ln|x|+K}$
$1+y = C|x|$
$2 = C$
$1+y = 2|x|$
$y = 2|x| - 1$ and $x < 0$
or
$y = -2x - 1$ and $x < 0$

7-7b

48. Consider the differential equation $\frac{dy}{dx} = (3 - y)\cos x$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(0) = 1$. The function f is defined for all real numbers.

(a) A portion of the slope field of the differential equation is given below. Sketch the solution curve through the point (0,1).

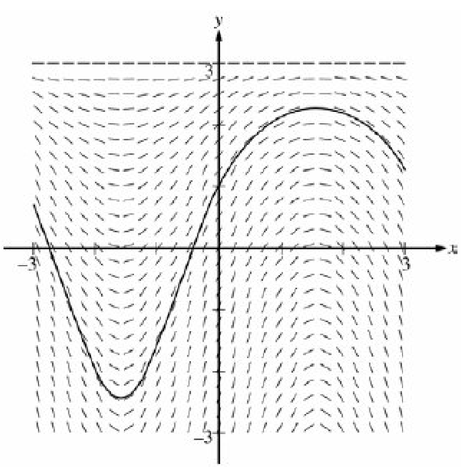


(b) Write an equation for the line tangent to the solution curve in part (a) at the point (0,1). Use the equation to approximate $f(0.2)$.

(c) Find $y = f(x)$, the particular solution to the differential equation with the initial condition $f(0) = 1$.

Part A

1 point is earned for solution curve



0	1

The student response earns all of the following points:

1 point is earned for solution curve

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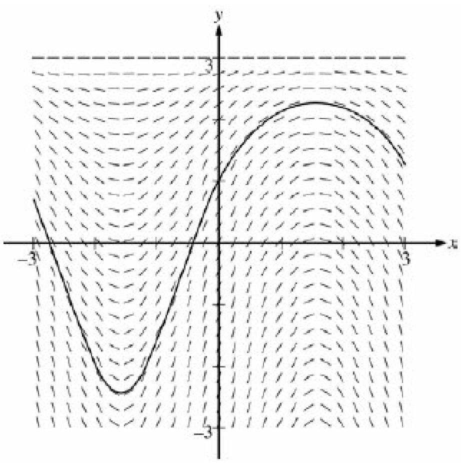

Part B

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left.\frac{dy}{dx}\right|_{(x,y)=(0,1)} = 2\cos 0 = 2$$

An equation for the tangent line is $y = 2x + 1$.

$f(0.2) \approx 2(0.2) + 1 = 1.4$

0	1	2

The student response earns all of the following points:

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left.\frac{dy}{dx}\right|_{(x,y)=(0,1)} = 2\cos 0 = 2$$

An equation for the tangent line is $y = 2x + 1$.

$f(0.2) \approx 2(0.2) + 1 = 1.4$

Part C

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

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$$\frac{dy}{dx} = (3 - y)\cos x$$
$$\int \frac{dy}{3 - y} = \int \cos x dx$$
$$-\ln|3 - y| = \sin x + C$$
$$-\ln 2 = \sin 0 + C \Rightarrow C = -\ln 2$$
$$-\ln|3 - y| = \sin x - \ln 2$$

Because $y(0) = 1$, $y < 3$, so $|3 - y| = 3 - y$

$$3 - y = 2e^{-\sin x}$$
$$y = 3 - 2e^{-\sin x}$$

Note: this solution is valid for all real numbers.

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

$$\frac{dy}{dx} = (3 - y)\cos x$$
$$\int \frac{dy}{3 - y} = \int \cos x dx$$
$$-\ln|3 - y| = \sin x + C$$
$$-\ln 2 = \sin 0 + C \Rightarrow C = -\ln 2$$
$$-\ln|3 - y| = \sin x - \ln 2$$

Because $y(0) = 1$, $y < 3$, so $|3 - y| = 3 - y$

$$3 - y = 2e^{-\sin x}$$
$$y = 3 - 2e^{-\sin x}$$

Note: this solution is valid for all real numbers.

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

7-7b

49. *Consider the differential equation* $\frac{dy}{dx} = \frac{3-x}{y}$.

(a) Let y = f(x) be the particular solution to the given differential equation for 1 < x < 5 such that the line y = -2 is tangent to the graph of f. Find the x-coordinate of the point of tangency, and determine whether f has a local maximum, local minimum, or neither at this point. Justify your answer.

(b) Let $y = g(x)$ be the particular solution to the given differential equation for $-2 < x < 8$, with the initial condition $g(6) = -4$. Find $y = g(x)$.

Part A

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$
$$\left.\frac{d^2y}{dx^2}\right|_{(3,-2)} = \left.\frac{-y-y'(3-x)}{y^2}\right|_{(3,-2)} = \frac{1}{2},$$

so f has a local minimum at this point.

or

Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$.Therefore f has a local minimum at $x = 3$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$
$$\left.\frac{d^2y}{dx^2}\right|_{(3,-2)} = \left.\frac{-y-y'(3-x)}{y^2}\right|_{(3,-2)} = \frac{1}{2},$$

so f has a local minimum at this point.

or

Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$.Therefore f has a local minimum at $x = 3$.

Part B

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $g(6) = -4$

1 point is earned for solving for y

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$y\,dy = (3 - x)dx$
$\frac{1}{2}y^2 = 3x - \frac{1}{2}x^2 + C$
$8 = 18 - 18 + C;\ C = 8$
$y^2 = 6x - x^2 + 16$
$y = -\sqrt{6x - x^2 + 16}$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivative of *dy* term

1 point is earned for the antiderivative of *dx* term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $g(6) = -4$

1 point is earned for solving for *y*

$y\,dy = (3 - x)dx$
$\frac{1}{2}y^2 = 3x - \frac{1}{2}x^2 + C$
$8 = 18 - 18 + C;\ C = 8$
$y^2 = 6x - x^2 + 16$
$y = -\sqrt{6x - x^2 + 16}$

7-7b

50. Consider the differential equation $\frac{dy}{dx} = \frac{3-x}{y}$.

Let $y = g(x)$ be the particular solution to the given differential equation for $-2 < x < 8$, with the initial condition $g(6) = -4$. Find $y = g(x)$.

Part B

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition g(6) = -4

1 point is earned for solves for *y*

$ydy = (3-x)dx$
$\frac{1}{2}y^2 = 3x - \frac{1}{2}x^2 + C$
$8 = 18 - 18 + C; C = 8$
$y^2 = 6x - x^2 + 16$
$y = -\sqrt{6x - x^2 + 16}$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition g(6) = -4

1 point is earned for solves for *y*

$ydy = (3-x)dx$
$\frac{1}{2}y^2 = 3x - \frac{1}{2}x^2 + C$
$8 = 18 - 18 + C; C = 8$
$y^2 = 6x - x^2 + 16$
$y = -\sqrt{6x - x^2 + 16}$

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51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

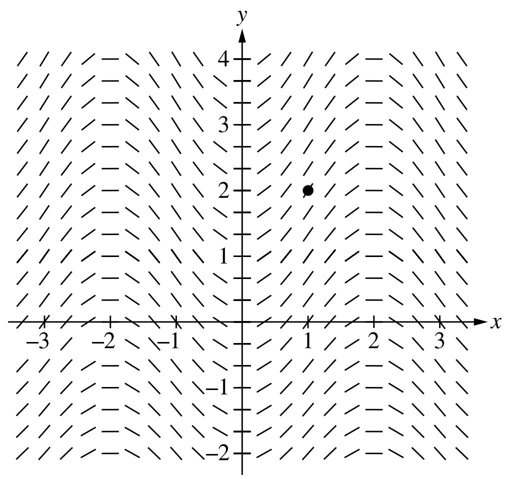
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}\sin\left(\frac{\pi}{2}x\right)\sqrt{y+7}$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(1) = 2$. The function f is defined for all real numbers.

(a) A portion of the slope field for the differential equation is given. Sketch the solution curve through the point $(1, 2)$.



(b) Write an equation for the line tangent to the solution curve in part (a) at the point $(1, 2)$. Use the equation to approximate $f(0.8)$.

(c) It is known that $f''(x) > 0$ for $-1 \le x \le 1$. Is the approximation found in part (b) an overestimate or an underestimate for $f(0.8)$? Give a reason for your answer.

(d) Use separation of variables to find $y = f(x)$, the particular solution to the differential equation $\frac{dy}{dx} = \frac{1}{2}\sin\left(\frac{\pi}{2}x\right)\sqrt{y+7}$ with the initial condition $f(1) = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The solution curve must pass through the point $(1, 2)$, extend reasonably close to the left and right edges of the square and have no obvious conflicts with the given slope lines.

Only portions of the solution curve within the given slope field are considered.

The solution curve must indicate $f(x) > 0$ for all points on the curve.

All local maximum/minimum points on the solution curve must occur at horizontal line segments in the slope field**.**

0	1 ✓

The student response accurately includes the criteria below.

☐ Solution curve

Solution:

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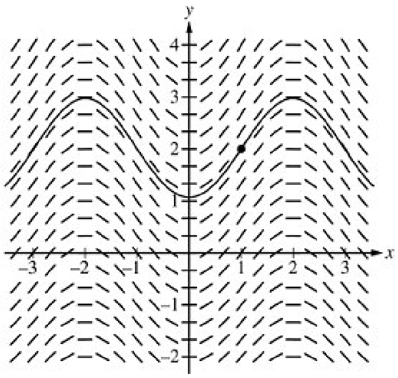

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The tangent line equation can be presented in any equivalent form.

An incorrect tangent line equation with a slope of $\frac{3}{2}$ is eligible to earn the second point for a consistent answer.

A response of only $2+\frac{3}{2}(0.8-1)$ earns the second point but not the first.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Tangent line equation
- [ ] Approximation

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\,y)=(1,\,2)}=\frac{1}{2}\cdot 3\cdot\ \sin\left(\frac{\pi}{2}\right)=\frac{3}{2}$$

An equation for the tangent line is $y=2+\frac{3}{2}(x-1)$.

$f\,(0.8)\approx 2+\frac{3}{2}(0.8-1)=1.7$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to

score the response.

The reason must include $f''(x)>0$, $f'(x)$ is increasing, or $f\,(x)$ is concave up.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

Because $f''(x)>0, f$ is concave up on $-1\leq x\leq 1$, the tangent line lies below the graph of $y=f\,(x)$ at $x=0.8$, and the approximation for $f\,(0.8)$ is an underestimate.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 5 points.

A response with no constant of integration can earn at most the first 3 points.

A response is eligible for the fourth point only if it has earned the first point and at least 1 of the 2 antiderivative points.

· Special case: The incorrect separation of $\sqrt{y+7}\ \ dy = \frac{1}{2}\sin\left(\frac{\pi}{2}x\right)\ dx$ does not earn the first point, is only eligible for the antiderivative point for $-\frac{1}{\pi}\cos\left(\frac{\pi}{2}x\right)$, and is eligible for the fourth point.

An eligible response earns the fourth point by correctly including the constant of integration in an equation and substituting 1 for x and 2 for y.

A response is eligible for the fifth point only if it has earned the first 4 points.

0	1	2	3	4	5

The student response accurately includes **all five** of the criteria below.

- ☐ Separation of variables
- ☐ One correct antiderivative
- ☐ The other correct antiderivative
- ☐ Constant of integration and uses initial condition
- ☐ Solves for y

Solution:

$$\int \frac{dy}{\sqrt{y+7}} = \int \frac{1}{2}\sin\left(\frac{\pi}{2}x\right) dx$$

$$2\sqrt{y+7} = -\frac{1}{\pi}\cos\left(\frac{\pi}{2}x\right) + C$$

$$f(1) = 2 \ \Rightarrow 2\sqrt{2+7} = -\frac{1}{\pi}\cos\left(\frac{\pi}{2}\cdot 1\right) + C$$

$$\Rightarrow\ 6 = -\frac{1}{\pi}\cos\left(\frac{\pi}{2}\right) + C \Rightarrow C = 6$$

$$\sqrt{y+7} = 3 - \frac{1}{2\pi}\cos\left(\frac{\pi}{2}x\right)$$

$$y = \left(3 - \frac{1}{2\pi}\cos\left(\frac{\pi}{2}x\right)\right)^2 - 7$$

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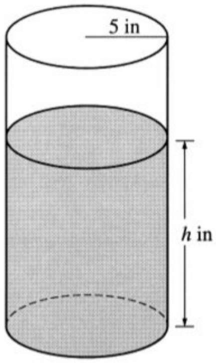
52.



A coffeepot has the shape of a cylinder with radius 5 inches, as shown in the figure above. Let h be the depth of the coffee in the pot, measured in inches, where h is a function of time t, measured in seconds. The volume V of coffee in the pot is changing at the rate of $-5\pi\sqrt{h}$ cubic inches per second. (The volume V of a cylinder with radius r and height h is $V = \pi r^2 h$.)

(a) Show that $\frac{dh}{dt} = -\frac{\sqrt{h}}{5}$.

(b) Given that $h = 17$ at time $t = 0$, solve the differential equation $\frac{dh}{dt} = -\frac{\sqrt{h}}{5}$ for h as a function of t.

(c) At what time t is the coffeepot empty?

Part A

1 point is earned for $\frac{dV}{dt} = -5\pi\sqrt{h}$

1 point is earned for computing $\frac{dV}{dt}$

1 point is earned for showing result

$$V = 25\pi h$$
$$\frac{dV}{dt} = 25\pi\frac{dh}{dt} = -5\pi\sqrt{h}$$
$$\frac{dh}{dt} = \frac{-5\pi\sqrt{h}}{25\pi} = -\frac{\sqrt{h}}{5}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for $\frac{dV}{dt} = -5\pi\sqrt{h}$

1 point is earned for computing $\frac{dV}{dt}$

1 point is earned for showing result

$$V = 25\pi h$$
$$\frac{dV}{dt} = 25\pi\frac{dh}{dt} = -5\pi\sqrt{h}$$
$$\frac{dh}{dt} = \frac{-5\pi\sqrt{h}}{25\pi} = -\frac{\sqrt{h}}{5}$$

Part B

1 point is earned for the correct separation of variables

$$\frac{dh}{dt} = -\frac{\sqrt{h}}{5}$$
$$\frac{1}{\sqrt{h}}dh = -\frac{1}{5}dt$$

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1 point is earned for the correct antiderivatives

1 point is earned for the correct constant of integration

$2\sqrt{h} = -\frac{1}{5}t + C$

1 point is earned for using initial condition $h = 17$ when $t = 0$

$2\sqrt{17} = 0 + C$

1 point is earned for correctly solving for h

$h = \left(-\frac{1}{10}t + \sqrt{17}\right)^2$

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for the correct separation of variables

$\frac{dh}{dt} = -\frac{\sqrt{h}}{5}$

$\frac{1}{\sqrt{h}}dh = -\frac{1}{5}dt$

1 point is earned for the correct antiderivatives

1 point is earned for the correct constant of integration

$2\sqrt{h} = -\frac{1}{5}t + C$

1 point is earned for using initial condition $h = 17$ when $t = 0$

$2\sqrt{17} = 0 + C$

1 point is earned for correctly solving for h

$h = \left(-\frac{1}{10}t + \sqrt{17}\right)^2$

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

Part C

1 point is earned for the correct answer

$(-\frac{1}{10}t + \sqrt{17})^2 = 0$

$t = 10\sqrt{17}$

0	1

The student response earns all of the following points:

1 point is earned for the correct answer

$(-\frac{1}{10}t + \sqrt{17})^2 = 0$

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$t = 10\sqrt{17}$

7-7b

53. Let f and g be functions that are differentiable for all real numbers x and that have the following properties.

(i) $f'(x) = f(x) - g(x)$

(ii) $g'(x) = g(x) - f(x)$

(iii) $f(0) = 5$

(iv) $g(0) = 1$

(a) Prove that $f(x) + g(x) = 6$ for all x.

(b) Find $f(x)$ and $g(x)$. Show your work.

Part A

1 point is earned for $f' + g' = 0$

1 point is earned for $f + g$ is constant

1 point is earned for uses $(f + g)(0)$

$f'(x) + g'(x) = f(x) - g(x) + g(x) - f(x) = 0$

so $f + g$ is constant.

$f(0) + g(0) = 6$, so $f(x) + g(x) = 6$

0	1	2	3

The student response earns all of the following points:

1 point is earned for $f' + g' = 0$

1 point is earned for $f + g$ is constant

1 point is earned for uses $(f + g)(0)$

$f'(x) + g'(x) = f(x) - g(x) + g(x) - f(x) = 0$

so $f + g$ is constant.

$f(0) + g(0) = 6$, so $f(x) + g(x) = 6$

Part B

1 point is earned for the correct differential equation for one function

1 point is earned for correctly separating variables

1 point is earned for the correct antidifferentiation

1 point is earned for correctly evaluating constant

1 point is earned for correctly finding g

1 point is earned for correctly finding f

Note: Max 1/6 if not working with DE for one function

7-7b

$f(x) = 6 - g(x)$ so
$g'(x) = g(x) - 6 + g(x) = 2g(x) - 6$
$\frac{dy}{dx} = 2y - 6;\ \frac{dy}{2y-6} = dx$
$\frac{1}{2}\ln\left|2y - 6\right| = x + C$
$\ln|2y - 6| = 2x + K$
$\left|2y - 6\right| = e^{2x+K}$
$2y - 6 = Ae^{2x}$
$x = 0 \Rightarrow y = 1$ so $-4 = A$
$2y = -4e^{2x} + 6$
$y = 3 - 2e^{2x} = g(x)$
$f(x) = 6 - g(x) = 3 + 2e^{2x}$

OR

1 point is earned for the correct equation in *f* - *g*

1 point is earned for the correct solution with constant

1 point is earned for correctly evaluating constant

1 point is earned for correctly using $f(x) + g(x) = 6$

1 point is earned for correctly finding *f*

1 point is earned for correctly finding *g*

$f(x) = 6 - g(x)$ so
$g'(x) = g(x) - 6 + g(x) = 2g(x) - 6$
$g'(x) - 2g(x) = -6$
$\frac{d}{dx}[g(x)e^{-2x}] = -6e^{-2x}$
$g(x)e^{-2x} = \int -6\mathrm{e}^{-2x}dx$
$= 3e^{-2x} + C$
$g(x) = 3 + Ce^{2x}$
$1 = g(0) = 3 + C;\ C = -2$
$g(x) = 3 - 2e^{2x}$
$f(x) = 6 - g(x) = 3 + 2e^{2x}$

OR

Student attempts solution with infinite series

1 point is earned for the correct constant terms in *f* and *g*

1 point is earned for the correct linear terms in *f* and *g*

1 point is earned for the correct quadratic term in *f*

1 point is earned for the correct quadratic term in *g*

1 point is earned for the correct general term for *f* or *g*

1 point is earned for the correct series sum to 6

7-7b

$f' - g' = 2f - 2g = 2(f - g)$, so
$f - g = Ae^{2x}$
$f(0) - g(0) = 4 = A$
$f(x) - g(x) = 4e^{2x}$
$\underline{f(x) + g(x) = 6}$
$2\,f(x) = 6 + 4e^{2x}$
$f(x) = 3 + 2e^{2x}$
$g(x) = 6 - f(x) = 3 - 2e^{2x}$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the correct differential equation for one function

1 point is earned for correctly separating variables

1 point is earned for the correct antidifferentiation

1 point is earned for correctly evaluating constant

1 point is earned for correctly finding g

1 point is earned for correctly finding f

Note: Max 1/6 if not working with DE for one function

$f(x) = 6 - g(x)$ so
$g'(x) = g(x) - 6 + g(x) = 2g(x) - 6$
$\frac{dy}{dx} = 2y - 6;\ \frac{dy}{2y-6} = dx$
$\frac{1}{2}\ln|2y - 6| = x + C$
$\ln|2y - 6| = 2x + K$
$|2y - 6| = e^{2x+K}$
$2y - 6 = Ae^{2x}$
$x = 0 \Rightarrow y = 1$ so $-4 = A$
$2y = -4e^{2x} + 6$
$y = 3 - 2e^{2x} = g(x)$
$f(x) = 6 - g(x) = 3 + 2e^{2x}$

OR

1 point is earned for the correct equation in f - g

1 point is earned for the correct solution with constant

1 point is earned for correctly evaluating constant

1 point is earned for correctly using $f(x) + g(x) = 6$

1 point is earned for correctly finding f

1 point is earned for correctly finding g

7-7b

$f(x) = 6 - g(x)$ so
$g'(x) = g(x) - 6 + g(x) = 2g(x) - 6$
$g'(x) - 2g(x) = -6$
$\frac{d}{dx}[g(x)e^{-2x}] = -6e^{-2x}$
$g(x)e^{-2x} = \int -6e^{-2x}dx$
$= 3e^{-2x} + C$
$g(x) = 3 + Ce^{2x}$
$1 = g(0) = 3 + C;\ C = -2$
$g(x) = 3 - 2e^{2x}$
$f(x) = 6 - g(x) = 3 + 2e^{2x}$

OR

Student attempts solution with infinite series

1 point is earned for the correct constant terms in f and g

1 point is earned for the correct linear terms in f and g

1 point is earned for the correct quadratic term in f

1 point is earned for the correct quadratic term in g

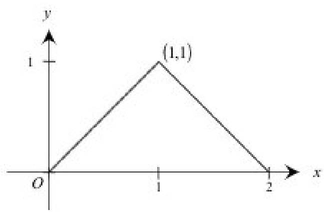
1 point is earned for the correct general term for f or g

1 point is earned for the correct series sum to 6

$f' - g' = 2f - 2g = 2(f - g)$, so
$f - g = Ae^{2x}$
$f(0) - g(0) = 4 = A$
$f(x) - g(x) = 4e^{2x}$
$\underline{f(x) + g(x) = 6}$
$2f(x) = 6 + 4e^{2x}$
$f(x) = 3 + 2e^{2x}$
$g(x) = 6 - f(x) = 3 - 2e^{2x}$

7-7b

54.



Note: This is the graph of the derivative of f, not the graph of f.

f' , the derivative of f. The domain of f is the set of all x such that $0 < x < 2$.

Given that $f(1)=0$, write an expression for $f(x)$ in terms of x.

Part B

1 point is earned for anti derivative for $x \leq 1$

1 point is earned for anti derivative for $x \geq 1$

2 points are earned for constants

$$f(x) = \begin{cases} \frac{x^2}{2} + C_1 & \text{if } 0 < x \leq 1 \\ 2x - \frac{x^2}{2} + C_2 & \text{if} 1 \leq x < 2 \end{cases}$$

$$0 = f(1) = \frac{1}{2} + C_1 \Rightarrow C_1 = -\frac{1}{2}$$

$$0 = f(1) = 2 - \frac{1}{2} + C_2 \Rightarrow C_2 = -\frac{3}{2}$$

$$f(x) = \begin{cases} \frac{x^2}{2} - \frac{1}{2} & \text{if } 0 < x \leq 1 \\ 2x - \frac{x^2}{2} - \frac{3}{2} & \text{if} 1 \leq x < 2 \end{cases}$$

or

$$f(x) = x - \frac{1}{2}(x-1)\left|x-1\right| - 1$$

0	1	2	3	4

The student response earns four of the following points:

1 point is earned for anti derivative for $x \leq 1$

1 point is earned for anti derivative for $x \geq 1$

2 points are earned for constants

7-7b

$$f(x)=\begin{cases}\frac{x^2}{2}+C_1 & \text{if } 0<x\le 1\\ 2x-\frac{x^2}{2}+C_2 & \text{if} 1\le x<2\end{cases}$$

$0=f(1)=\frac{1}{2}+C_1 \Rightarrow C_1=-\frac{1}{2}$

$0=f(1)=2-\frac{1}{2}+C_2 \Rightarrow C_2=-\frac{3}{2}$

$$f(x)=\begin{cases}\frac{x^2}{2}-\frac{1}{2} & \text{if } 0<x\le 1\\ 2x-\frac{x^2}{2}-\frac{3}{2} & \text{if} 1\le x<2\end{cases}$$

or

$f(x)=x-\frac{1}{2}(x-1)\left|x-1\right|-1$

7-7b

55. Let f be the function that is defined for all real numbers x and that has the following properties.

(i) $f''(x) = 24x - 18$

(ii) $f'(1) = -6$

(iii) $f(2) = 0$

(a) Find each x such that the line tangent to the graph of f at $(x, f(x))$ is horizontal.

(b) Write an expression for $f(x)$.

(c) Find the average value of f on the interval $1 \le x \le 3$.

Part A

1 point is earned for antiderivative with or without C

1 point is earned for $f'(1) = -6$ for student's $f'(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$$f'(x) = 12x^2 - 18x + C$$
$$f'(1) = -6 = 12 - 18 + C$$
$$\therefore C = 0$$
$$f'(x) = 6x(2x - 3) = 0$$
$$\boxed{x = 0, \tfrac{3}{2}}$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for antiderivative with or without C

1 point is earned for $f'(1) = -6$ for student's $f'(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$$f'(x) = 12x^2 - 18x + C$$
$$f'(1) = -6 = 12 - 18 + C$$
$$\therefore C = 0$$
$$f'(x) = 6x(2x - 3) = 0$$
$$\boxed{x = 0, \tfrac{3}{2}}$$

Part B

1 point is earned for antiderivative with or without C

1 point is earned for equating student's $f(2)$ and zero

1 point is earned for solving the resulting equation

$$f(x) = 4x^3 - 9x^2 + C$$
$$f(2) = 0 = 32 - 36 + C$$
$$\therefore C = 4$$
$$\boxed{f(x) = 4x^3 - 9x^2 + 4}$$

7-7b

0	1	2	3

The student response earns all of the following points:

1 point is earned for antiderivative with or without C

1 point is earned for equating student's $f(2)$ and zero

1 point is earned for solving the resulting equation

$$f(x) = 4x^3 - 9x^2 + C$$
$$f(2) = 0 = 32 - 36 + C$$
$$\therefore C = 4$$
$$\boxed{f(x) = 4x^3 - 9x^2 + 4}$$

Part C

1 point is earned for setting up integral with students $f(x)$ from part (b)

1 point is earned for the antiderivative and evaluation

Note: 0/2 if integrand is not student's $f(x)$

$$\frac{1}{3-1}\int_1^3 4x^3 - 9x^2 + 4 \text{ dx}$$
$$= \frac{1}{2}\left[x^4 - 3x^3 + 4x\right]_1^3$$
$$= \frac{1}{2}[(81 - 81 + 12) - (1 - 3 + 4)]$$
$$= \boxed{5}$$

0	1	2

The student response earns all of the following points:

1 point is earned for setting up integral with students $f(x)$ from part (b)

1 point is earned for the antiderivative and evaluation

Note: 0/2 if integrand is not student's $f(x)$

$$\frac{1}{3-1}\int_1^3 4x^3 - 9x^2 + 4 \text{ dx}$$
$$= \frac{1}{2}\left[x^4 - 3x^3 + 4x\right]_1^3$$
$$= \frac{1}{2}[(81 - 81 + 12) - (1 - 3 + 4)]$$
$$= \boxed{5}$$

7-7b

56. The number of gallons, *P(t)*, of a pollutant in a lake changes at the rte $P'(t) = 1 - 3e^{-0.2\sqrt{t}}$ gallons per day, where *t* is measured in days. There are 50 gallons of the pollutant in the lake at time $t = 0$. The lake is considered to be safe when it contains 40 gallons or less of pollutant.

Is the lake safe when the number of gallons of pollutant is at its minimum? Justify your answer.

Part C

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the conclusion with reason based on integral of *P′(t)*

$$P\left(30.174\right) = 50 + \int_0^{30.174}\left(1 - 3e^{-0.2\sqrt{t}}\right)dt$$

= 35.104 < 40, so the lake is safe.

0	1	2	3

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the conclusion with reason based on integral of *P′(t)*

$$P\left(30.174\right) = 50 + \int_0^{30.174}\left(1 - 3e^{-0.2\sqrt{t}}\right)dt$$

= 35.104 < 40, so the lake is safe.

57. If dy/dx=y sec^{2}x and *y*=5 when *x*=0, then *y*=

(A) $e^{\tan x} + 4$

(B) $e^{\tan x} + 5$

(C) $5e^{\tan x}$

(D) $\tan x + 5$

(E) $\tan x + 5e^{x}$

7-7b

58. Oil is being pumped continuously from a certain oil well at a rate proportional to the amount of oil left in the well; that is,dy/dt=ky , where *y* is the amount of oil left in the well at any time *t*. Initially there were 1,000,000 gallons of oil in the well, and 6 years later there were 500,000 gallons remaining. It will no longer be profitable to pump oil when there are fewer than 50,000 gallons remaining.

Write an equation for *y*, the amount of oil remaining in the well at any time *t*.

Part A

2 points are earned for $y = Ce^{kt}$ **or** 1 point for each: $\ln|y| = kt + C, y = e^{kt+C}$

1 point is earned for $C = 10^6$

1 point is earned for $k = -\frac{\ln 2}{6}$

1 point is earned for answer

*Max 2/5 If *y* is not an exponential

$$\frac{dy}{dt} = ky$$

$$y = Ce^{kt} \quad or \quad \begin{cases} \frac{dy}{y} = kdt \\ \ln|y| = kt + C_1 \\ y = e^{kt+C_1} \end{cases}$$

$$t = 0 \Rightarrow C \Rightarrow 10^6,\ C_1 = \ln 10^6$$

$$\therefore y = 10^6 e^{kt}$$

$$t = 6 \Rightarrow \frac{1}{2} = e^{6k}$$

$$\therefore k = -\frac{\ln 2}{6}$$

$$y = 10^6 e^{\frac{-t}{6}\ln 2} = 10^6 \cdot 2^{\frac{-t}{6}}$$

0	1	2	3	4	5

The student response earns five of the following points:

2 points are earned for $y = Ce^{kt}$ **or** 1 point for each: $\ln|y| = kt + C, y = e^{kt+C}$

1 point is earned for $C = 10^6$

1 point is earned for $k = -\frac{\ln 2}{6}$

1 point is earned for answer

*Max 2/5 If *y* is not an exponential

$$\frac{dy}{dt} = ky$$

$$y = Ce^{kt} \quad or \quad \begin{cases} \frac{dy}{y} = kdt \\ \ln|y| = kt + C_1 \\ y = e^{kt+C_1} \end{cases}$$

$$t = 0 \Rightarrow C \Rightarrow 10^6,\ C_1 = \ln 10^6$$

$$\therefore y = 10^6 e^{kt}$$

$$t = 6 \Rightarrow \frac{1}{2} = e^{6k}$$

$$\therefore k = -\frac{\ln 2}{6}$$

$$y = 10^6 e^{\frac{-t}{6}\ln 2} = 10^6 \cdot 2^{\frac{-t}{6}}$$

7-7b

59. Which of the following is the solution to the differential equation $\frac{dy}{dx} = \frac{4x}{y}$, where $y(2) = -2$?

(A) $y = 2x$ for $x > 0$

(B) $y = 2x - 6$ for $x \neq 3$

(C) $y = -\sqrt{4x^2 - 12}$ for $x > \sqrt{3}$ ✓

(D) $y = \sqrt{4x^2 - 12}$ for $x > \sqrt{3}$

(E) $y = -\sqrt{4x^2 - 6}$ for $x > \sqrt{1.5}$

60. Which of the following is the solution to the differential equation $\frac{dy}{dx} = \frac{x^2}{y}$ with the initial condition $y(3) = -2$?

(A) $y = 2e^{-9+x^3/3}$

(B) $y = -2e^{-9+x^3/3}$

(C) $y = \sqrt{\frac{2x^3}{3}}$

(D) $y = \sqrt{\frac{2x^3}{3} - 14}$

(E) $y = -\sqrt{\frac{2x^3}{3} - 14}$ ✓

61. Which of the following is the solution to the differential equation $\frac{dy}{dx} = 2\sin x$ with the initial condition $y(\pi) = 1$?

(A) $y = 2\cos x + 3$

(B) $y = 2\cos x - 1$

(C) $y = -2\cos x + 3$

(D) $y = -2\cos x + 1$

(E) $y = -2\cos x - 1$ ✓

62. If $\frac{dy}{dx} = y\sec^2 x$ and $y = 5$ when $x = 0$, then $y =$

(A) $e^{\tan x} + 4$

(B) $e^{\tan x} + 5$

(C) $5e^{\tan x}$ ✓

(D) $\tan x + 5$

(E) $\tan x + 5e^x$

7-7b

63. Let $v(t)$ be the velocity, in feet per second, of a skydiver at time t seconds, $t \geq 0$. After her parachute opens, her velocity satisfies the differential equation $\frac{dv}{dt} = -2v - 32$, with initial condition $v(0) = -50$.

Use separation of variables to find an expression for v in terms of t, where t is measured in seconds.

Part A

Solution: Max 6/6

1 point is earned for correctly separating variables

$$\frac{dv}{dt} = -2v - 32 = -2(v + 16)$$

1 point is earned for the correct antiderivative of dv side

0/1 if not $\int \frac{dv}{\alpha v + b}, \alpha \neq 0$

$$\frac{dv}{v+16} = -2dt$$

1 point is earned for the correct antiderivative of dt side

$$\ln|v + 16| = -2t + A$$

1 point is earned for the correct constant of integration

$$\left|v + 16\right| = e^{-2t+A} = e^A e^{-2t}$$

1 point is earned for correctly using initial condition v(0)=-50

$$v + 16 = Ce^{-2t}$$
$$-50 + 16 = Ce^0; C = -34$$

1 point is earned for correctly solving for $v(t)$

0/1 if not solving $\frac{dv}{\alpha v+b} = kdt$ where a, b, k nonzero

0/1 if no constant of integration

0/6 if variables not separated

$$v = -34e^{-2t} - 16$$

OR

Integrating Factor Solution: Max 5/6

1 point is earned for correct use of integrating factor

$$v'(t) = -2v(t) - 32$$
$$e^{2t}(v'(t) + 2v(t)) = -32e^{2t}$$
$$\frac{d}{dt}(v(t)e^{2t}) = -32e^{2t}$$
$$v(t)e^{2t} = \int -32e^{2t}dt = -16e^{2t} + C$$
$$v(t) = -16 + Ce^{-2t}$$
$$-50 = -16 + C; C = -34$$

4 points are earned for the correct solution

$$v(t) = -16 - 34e^{-2t}$$

0	1	2	3	4	5	6

The student response earns six of the following points:

7-7b

Solution: Max 6/6

1 point is earned for correctly separating variables

$\frac{dv}{dt} = -2v - 32 = -2(v+16)$

1 point is earned for the correct antiderivative of *dv* side

0/1 if not $\int \frac{dv}{\alpha v + b}, \alpha \neq 0$

$\frac{dv}{v+16} = -2dt$

1 point is earned for the correct antiderivative of *dt* side

$\ln|v+16| = -2t + A$

1 point is earned for the correct constant of integration

$\left|v+16\right| = e^{-2t+A} = e^{A}e^{-2t}$

1 point is earned for correctly using initial condition $v(0) = -50$

$v + 16 = Ce^{-2t}$
$-50 + 16 = Ce^{0}; C = -34$

1 point is earned for correctly solving for $v(t)$

0/1 if not solving $\frac{dv}{\alpha v+b} = kdt$ where a, b, k nonzero

0/1 if no constant of integration

0/6 if variables not separated

$v = -34e^{-2t} - 16$

OR

Integrating Factor Solution: Max 5/6

1 point is earned for correct use of integrating factor

$v'(t) = -2v(t) - 32$
$e^{2t}(v'(t) + 2v(t)) = -32e^{2t}$
$\frac{d}{dt}(v(t)e^{2t}) = -32e^{2t}$
$v(t)e^{2t} = \int -32e^{2t}dt = -16e^{2t} + C$
$v(t) = -16 + Ce^{-2t}$
$-50 = -16 + C; C = -34$

4 points are earned for the correct solution

$v(t) = -16 - 34e^{-2t}$

7-7b

64. A particle moves along the y-axis with velocity given by $v(t) = t\sin(t^2)$ for $t \geq 0$.
Given that $y(t)$ is the position of the particle at time t and that $y(0) = 3$, find $y(2)$.

Part C

1 point is earned for: $y(t) = \int v(t)dt$

1 point is earned for: $y(t) = -\frac{1}{2}\cos t^2 + C$

1 point is earned for: $y(2)$

$y(t) = \int v(t)dt$
$= \int t\sin t^2 dt = -\frac{\cos t^2}{2} + C$
$y(0) = 3 = -\frac{1}{2} + C \Rightarrow C = \frac{7}{2}$
$y(1) = -\frac{1}{2}\cos t^2 + \frac{7}{2}$
$y(2) = -\frac{1}{2}\cos 4 + \frac{7}{2} = 3.826, or, 3.827$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $y(t) = \int v(t)dt$

1 point is earned for: $y(t) = -\frac{1}{2}\cos t^2 + C$

1 point is earned for: $y(2)$

$y(t) = \int v(t)dt$
$= \int t\sin t^2 dt = -\frac{\cos t^2}{2} + C$
$y(0) = 3 = -\frac{1}{2} + C \Rightarrow C = \frac{7}{2}$
$y(1) = -\frac{1}{2}\cos t^2 + \frac{7}{2}$
$y(2) = -\frac{1}{2}\cos 4 + \frac{7}{2} = 3.826, or, 3.827$

7-7b

Water is pumped into an underground tank at a constant rate of 8 gallons per minute. Water leaks out of the tank at the rate of $\sqrt{t+1}$ gallons per minute, for $0 \le t \le 120$ minutes. At time $t = 0$, the tank contains 30 gallons of water.

65. How many gallons of water are in the tank at time $t = 3$ minutes?

Part B

1 point is earned for the answer

$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$

0	1

The student response earns all of the following points:

1 point is earned for the answer

$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$

7-7b

66. Write an expression for *A(t)*, the total number of gallons of water in the tank at *t*.

Part C

Method 1:

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1}dx$

- or -

Method 2:

1 point earned for antiderivative with *C*

1 point is earned for the answer

Method 1:
$$A(t) = 30 + \int_0^t \left(8 - \sqrt{x+1}\right)dx$$
$$= 30 + 8t - \int_0^t \sqrt{x+1}dx$$
$-or-$

Method 2:
$$\frac{dA}{dt} = 8 - \sqrt{t+1}$$
$$A(t) = 8t - \tfrac{2}{3}(t+1)^{3/2} + C$$
$$30 = 8(0) - \tfrac{2}{3}(0+1)^{3/2} + C; C = \frac{92}{3}$$
$$A(t) = 8t - \tfrac{2}{3}(t+1)^{3/2} + \frac{92}{3}$$

0	1	2	3	4 ✓

The student response earns all of the following points:

Method 1:

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1}dx$

- or -

Method 2:

1 point earned for antiderivative with *C*

1 point is earned for the answer

Method 1:
$$A(t) = 30 + \int_0^t \left(8 - \sqrt{x+1}\right)dx$$
$$= 30 + 8t - \int_0^t \sqrt{x+1}dx$$
$-or-$

7-7b

Method 2:

$$\frac{dA}{dt} = 8 - \sqrt{t+1}$$
$$A(t) = 8t - \tfrac{2}{3}(t+1)^{3/2} + C$$
$$30 = 8(0) - \tfrac{2}{3}(0+1)^{3/2} + C; C = \frac{92}{3}$$
$$A(t) = 8t - \tfrac{2}{3}(t+1)^{3/2} + \frac{92}{3}$$

67. If $dy/dx=2y^2$ and if y=-1 when x=1, then when x=2, y=

(A) −2/3

(B) −1/3 ✓

(C) 0

(D) 1/3

(E) 2/3

7-7b

68. Consider the differential equation $\frac{dy}{dx} = x^2\,(y-1)$.

Find the particular solution $y = f(x)$ to the given differential equation with initial condition $f(0) = 3$.

Part C

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for the uses initial condition

1 point is earned for solves for y

0/1 if y is not exponential

$\frac{1}{y-1}dy = x^2dx$
$\ln|y-1| = \frac{1}{3}x^3 + C$
$|y-1| = e^Ce^{\frac{1}{3}x^3}$
$y-1 = Ke^{\frac{1}{3}x^3}, K = \pm e^C$
$2 = Ke^0 = K$
$y = 1 + 2e^{\frac{1}{3}x^3}$

0	1	2	3	4	5	6 ✓

The student response earns all of the following points:

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for the uses initial condition

1 point is earned for solves for y

0/1 if y is not exponential

$\frac{1}{y-1}dy = x^2dx$
$\ln|y-1| = \frac{1}{3}x^3 + C$
$|y-1| = e^Ce^{\frac{1}{3}x^3}$
$y-1 = Ke^{\frac{1}{3}x^3}, K = \pm e^C$
$2 = Ke^0 = K$
$y = 1 + 2e^{\frac{1}{3}x^3}$

69. If $\frac{dy}{dx} = \sin x\cos^2 x$ and if y=0 when $x = \frac{\pi}{2}$, what is the value of y when x=0 ?

(A) -1
(B) $-\frac{1}{3}$ ✓
(C) 0
(D) $\frac{1}{3}$
(E) 1

7-7b

70. A particle moves along the x-axis so that at any time $t > 0$ its velocity is given by $v(t) = t \ln t - t$. At time $t = 1$, the position of the particle is $x(1) = 6$.

(a) Write an expression for the acceleration of the particle.

(b) For what values of t is the particle moving to the right?

(c) What is the minimum velocity of the particle? Show the analysis that leads to your conclusion.

(d) Write an expression of the position $x(t)$ of the particle.

Part A

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$

0	1 ✓

The student earns all of the following points:

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$

Part B

1 point is earned for correctly setting $t \ln t - t > 0$

$v(t) = t \ln t - t > 0$

$= t\,(\ln t - 1) > 0$

1 point is earned for the correct answer

$t > e$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for correctly setting $t \ln t - t > 0$

$v(t) = t \ln t - t > 0$

$= t\,(\ln t - 1) > 0$

1 point is earned for the correct answer

$t > e$

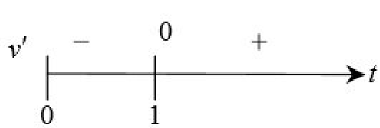
Part C

1 point is earned for correctly solving $v'(t) = 0$

$v'(t) = \ln t = 0$

$t = 1$

1 point is earned for the correct analysis



7-7b

1 point is earned for the correct answer

minimum velocity is $v(1) = -1$

Max 1/3 if student's v' does not involve ln

0	1	2	3

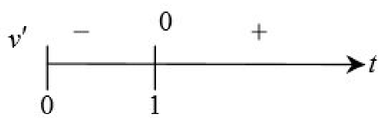
The student response earns all of the following points:

1 point is earned for correctly solving $v'(t) = 0$

$v'(t) = \ln t = 0$

$t = 1$

1 point is earned for the correct analysis



1 point is earned for the correct answer

minimum velocity is $v(1) = -1$

Max 1/3 if student's v' does not involve ln

Part D

2 points are earned for the correct antiderivative of $t \ln t - t$

<-2>: error in $\int$ -by-parts

<-1>: any other error

$$\int t\ln t - t\,dt$$
$$= \left(\tfrac{1}{2}t^2\ln t - \tfrac{1}{4}t^2\right) - \tfrac{1}{2}t^2 + C$$
$$= \tfrac{1}{2}t^2\ln t - \tfrac{3}{4}t^2 + C;$$
$$6 = x(1) = 0 - \tfrac{3}{4} + C$$

1 point is earned for correctly solving for C

$$C = \frac{27}{4};$$
$$x(t) = \tfrac{1}{2}t^2\ln t - \tfrac{3}{4}t^2 + \frac{27}{4}$$

0/3 if not antidifferentiating $t \ln t - t$

0	1	2	3

The student response earns all of the following points:

2 points are earned for the correct antiderivative of $t \ln t - t$

<-2>: error in $\int$ -by-parts

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<-1>: any other error

$$\int t\ln t - t\,dt$$
$$= \left(\tfrac{1}{2}t^2\ln t - \tfrac{1}{4}t^2\right) - \tfrac{1}{2}t^2 + C$$
$$= \tfrac{1}{2}t^2\ln t - \tfrac{3}{4}t^2 + C;$$
$$6 = x(1) = 0 - \tfrac{3}{4} + C$$

1 point is earned for correctly solving for C

$$C = \frac{27}{4};$$
$$x(t) = \tfrac{1}{2}t^2\ln t - \tfrac{3}{4}t^2 + \frac{27}{4}$$

0/3 if not antidifferentiating $t \ln t$ - t

7-7b

71. At the beginning of 2010, a landfill contained 1400 tons of solid waste. The increasing function W models the total amount of solid waste stored at the landfill. Planners estimate that W will satisfy the differential equation $\frac{dW}{dt} = \frac{1}{25}(W - 300)$ for the next 20 years. W is measured in tons, and t is measured in years from the start of 2010.

(a) Use the line tangent to the graph of W at $t = 0$ to approximate the amount of solid waste that the landfill contains at the end of the first 3 months of 2010 (time $t = \frac{1}{4}$).

(b) Find d^2W/dt^2 in terms of W. Use d^2W/dt^2 to determine whether your answer in part (a) is an underestimate or an overestimate of the amount of solid waste that the landfill contains at time $t = 1/4$.

(c) Find the particular solution $W = W(t)$ to the differential equation $\frac{dW}{dt} = \frac{1}{25}(W - 300)$ with initial condition $W(0) = 1400$.

Part A

1 point is earned for $\frac{dW}{dt}$ at $t = 0$

1 point is earned for the answer

$$\left.\frac{dW}{dt}\right|_{t=0} = \frac{1}{25}(W(0) - 300) = \frac{1}{25}(1400 - 300) = 44$$

The tangent line is $y = 1400 + 44t$

$$W\left(\frac{1}{4}\right) \approx 1400 + 44\left(\frac{1}{4}\right) = 1411 \text{ tons}$$

0	1	2

The student response earns all of the following points:

1 point is earned for $\frac{dW}{dt}$ at $t = 0$

1 point is earned for the answer

$$\left.\frac{dW}{dt}\right|_{t=0} = \frac{1}{25}(W(0) - 300) = \frac{1}{25}(1400 - 300) = 44$$

The tangent line is $y = 1400 + 44t$

$$W\left(\frac{1}{4}\right) \approx 1400 + 44\left(\frac{1}{4}\right) = 1411 \text{ tons}$$

Part B

1 point is earned for $\frac{d^2W}{dt^2}$

1 point is earned for the answer with reason

$$\frac{d^2W}{dt^2} = \frac{1}{25}\frac{dW}{dt} = \frac{1}{625}(W - 300) \text{ and } W \geq 1400$$

7-7b

Therefore $\frac{d^2W}{dt^2} > 0$ on the interval $0 \le t \le \frac{1}{4}$.

The answer in part(a) is an underestimate.

0	1	2

The student response earns all of the following points:

1 point is earned for $\frac{d^2W}{dt^2}$

1 point is earned for the answer with reason

$\frac{d^2W}{dt^2} = \frac{1}{25}\frac{dW}{dt} = \frac{1}{625}(W - 300)$ and $W \ge 1400$

Therefore $\frac{d^2W}{dt^2} > 0$ on the interval $0 \le t \le \frac{1}{4}$.

The answer in part(a) is an underestimate.

Part C

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for W

$$\frac{dW}{dt} = \frac{1}{25}(W - 300)$$
$$\int \frac{1}{W - 300} dW = \int \frac{1}{25} dt$$
$$\ln|W - 300| = \frac{1}{25}t + C$$
$$\ln(1400 - 300) = \frac{1}{25}(0) + C \Rightarrow \ln(1100) = C$$
$$W - 300 = 1100e^{\frac{1}{25}t}$$
$$W(t) = 300 + 1100e^{\frac{1}{25}t},\ 0 \le t \le 20$$

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

7-7b

1 point is earned for solving for W

$$\frac{dW}{dt} = \frac{1}{25}(W - 300)$$
$$\int \frac{1}{W-300} dW = \int \frac{1}{25} dt$$
$$\ln|W - 300| = \frac{1}{25}t + C$$
$$\ln(1400 - 300) = \frac{1}{25}(0) + C \Rightarrow \ln(1100) = C$$
$$W - 300 = 1100e^{\frac{1}{25}t}$$
$$W(t) = 300 + 1100e^{\frac{1}{25}t},\ 0 \le t \le 20$$

7-7b

72. Let $P(t)$ represent the number of wolves in a population at time t years, when $t \geq 0$. The population $P(t)$ is increasing at a rate directly proportional to 800 - $P(t)$, where the constant of proportionality is k.

(a) If $P(0) = 500$, find $P(t)$ in terms of t and k.

(b) If $P(2) = 700$, find k.

(c) Find $\lim\limits_{t \to \infty} P(t)$.

Part A

1 point is earned for differential equation

1 point is earned if separates variables

2 points are earned for anti differentiation

1 point is **deducted** for errors

1 points is earned for solving for constant

1 point is earned if solves for $p(t)$

Note: If no differential equation solved, then not eligible for last two points

$$P'(t) = k(800 - P(t))$$
$$\frac{dP}{800-P} = k\ dt$$
$$-\ln|800 - P| = kt + C_0$$
$$|800 - P| = C_1 e^{-kt}$$
$$800 - 500 = C_1 e^0$$
$$C_1 = 300$$
$$\therefore\ P(t) = 800 - 300e^{-kt}$$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for differential equation

1 point is earned if separates variables

2 points are earned for anti differentiation

1 point is **deducted** for errors

1 points is earned for solving for constant

1 point is earned if solves for $p(t)$

Note: If no differential equation solved, then not eligible for last two points

$$P'(t) = k(800 - P(t))$$
$$\frac{dP}{800-P} = k\ dt$$
$$-\ln|800 - P| = kt + C_0$$
$$|800 - P| = C_1 e^{-kt}$$
$$800 - 500 = C_1 e^0$$
$$C_1 = 300$$
$$\therefore\ P(t) = 800 - 300e^{-kt}$$

Part B

7-7b

1 point is earned for setting $P(2)$ equal to 700

1 point is earned for solving for k

$$P(2) = 700 = 800 - 300e^{-2k}$$
$$k = \frac{\ln 3}{2} \approx 0.549$$

0	1	2

The student response earns all of the following points:

1 point is earned for setting $P(2)$ equal to 700

1 point is earned for solving for k

$$P(2) = 700 = 800 - 300e^{-2k}$$
$$k = \frac{\ln 3}{2} \approx 0.549$$

Part C

1 point is earned for limits

$$\lim_{t\to\infty}\left(800 - 300e^{-\frac{\ln 3}{2}t}\right) = 800$$

0	1

The student response earns all of the following points:

1 point is earned for limits

$$\lim_{t\to\infty}\left(800 - 300e^{-\frac{\ln 3}{2}t}\right) = 800$$

7-7b

73. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

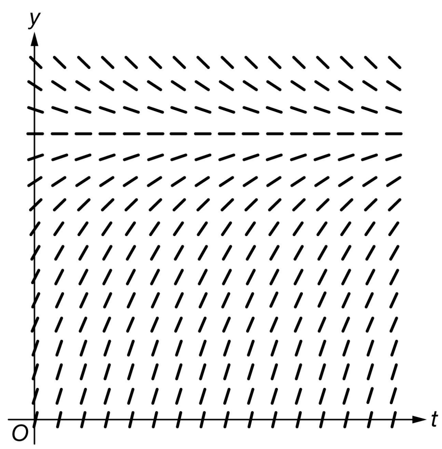
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A medication is administered to a patient. The amount, in milligrams, of the medication in the patient at time t hours is modeled by a function $y = A(t)$ that satisfies the differential equation $\frac{dy}{dt} = \frac{12-y}{3}$. At time $t = 0$ hours, there are 0 milligrams of the medication in the patient.

(a) A portion of the slope field for the differential equation $\frac{dy}{dt} = \frac{12-y}{3}$ is given below. Sketch the solution curve through the point $(0, 0)$.



(b) Using correct units, interpret the statement $\lim_{t\to\infty} A(t) = 12$ in the context of this problem.

(c) Use separation of variables to find $y = A(t)$, the particular solution to the differential equation $\frac{dy}{dt} = \frac{12-y}{3}$ with initial condition $A(0) = 0$.

(d) A different procedure is used to administer the medication to a second patient. The amount, in milligrams, of the medication in the second patient at time t hours is modeled by a function $y = B(t)$ that satisfies the differential equation $\frac{dy}{dt} = 3 - \frac{y}{t+2}$. At time $t = 1$ hour, there are 2.5 milligrams of the medication in the second patient. Is the rate of change of the amount of medication in the second patient increasing or decreasing at time $t = 1$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the solution curve must pass through the point $(0, 0)$, be generally increasing and concave down, and approach the horizontal asymptote from below as t increases. The point is not earned if two or more solution curves are presented.

0	1

The student response accurately includes the criteria below.

☐ Solution curve

Solution:

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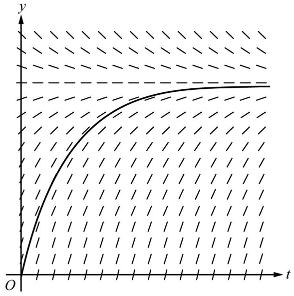


Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the interpretation must include “medication in the patient,” “approaches 12,” and units (milligrams), or their equivalents.

0	1

The student response accurately includes the criteria below.

- [ ] Interpretation

Solution:

Over time the amount of medication in the patient approaches 12 milligrams.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{dy}{12-y} = 3\ dt$ is a bad separation and does not earn the first point. However, this response is eligible for the second and third points. It cannot earn the fourth point.

Absolute value bars are not required in this part.

A response that correctly separates to $\frac{3\ dy}{12-y} = dt$ but then incorrectly simplifies to $\frac{dy}{4-y} = dt$ earns the first point (for the initial correct separation), is eligible for the second point (for $-\ln|4-y| = t$, with or without $+C$), but is not eligible for the third or fourth points.

$+\ln|12-y| = \frac{t}{3}$ (with or without $+C$) does not earn the second point and is not eligible for the fourth point; $+\ln|12-y| = \frac{t}{3} + C$ is eligible for the third point

In all other cases, the points are earned consecutively—the second point cannot be earned without the first, the third without the second, etc.

0	1	2	3	4

The student response accurately includes **all four** of the criteria below.

- [ ] Separation of variables
- [ ] Antiderivatives
- [ ] Constant of integration and uses initial condition
- [ ] Solves for y

Solution:

$$\frac{dy}{dt} = \frac{12-y}{3} \Rightarrow \frac{dy}{12-y} = \frac{dt}{3}$$

7-7b

$$\int \frac{dy}{12-y} = \int \frac{dt}{3} \Rightarrow -\ln|12-y| = \tfrac{t}{3} + C$$

$\ln|12-y| = -\frac{t}{3} - C \Rightarrow |12-y| = e^{-t/3-C}$
$\Rightarrow y = 12 + Ke^{-t/3}$

$0 = 12 + K \Rightarrow K = -12$

$y = A(t) = 12 - 12e^{-t/3}$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for correctly applying the quotient rule to $\frac{y}{t+2}$ or applying the product rule to $y(t+2)^{-1}$. Errors in differentiating the constant, 3, or handling the sign of the second term of $\frac{dy}{dt}$ will result in not earning the second point.

The second point cannot be earned unless the second derivative $\frac{d^2y}{dt^2}$ is correct.

For the second point it is sufficient to state the sign of $B''(1)$ is negative with supporting work. If a value is declared for $B''(1)$, it must be correct in order to earn the second point.

Eligibility for the third point: An attempt at using the quotient rule (or product rule) to find $B''(1)$. In this case the third point will be earned for a consistent conclusion based on the declared value (or sign) of $B''(1)$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Quotient rule
- ☐ $B''(1) < 0$
- ☐ Answer with reason

Solution:

$$\frac{dy}{dt} = 3 - \frac{y}{t+2} \Rightarrow \frac{d^2y}{dt^2} = (-1)\frac{\frac{dy}{dt}(t+2) - y}{(t+2)^2}$$

$$B'(1) = 3 - \frac{B(1)}{3} = 3 - \frac{2.5}{3} = \frac{6.5}{3}$$

$$B''(1) = -\frac{B'(1)\cdot 3 - B(1)}{3^2} = -\frac{6.5-2.5}{9} = -\frac{4}{9} < 0$$

The rate of change of the amount of medication is decreasing at time $t = 1$ because $B''(1) < 0$ and $\frac{d^2y}{dt^2}$ is continuous in an interval containing $t = 1$.

7-7b

74. For $0 \le t \le 31$, the rate of change of the number of mosquitoes on Tropical Island at time t days is modeled by $R(t) = 5\sqrt{t}\cos\left(\frac{t}{5}\right)$ mosquitoes per day. There are 1000 mosquitoes on Tropical Island at time $t = 0$.

(a) Show that the number of mosquitoes is increasing at time $t = 6$.

(b) At time $t = 6$, is the number of mosquitoes increasing at an increasing rate, or is the number of mosquitoes increasing at a decreasing rate? Give a reason for your answer.

(c) According to the model, how many mosquitoes will be on the island at time $t = 31$? Round your answer to the nearest whole number.

(d) To the nearest whole number, what is the maximum number of mosquitoes for $0 \le t \le 31$? Show the analysis that leads to your conclusion.

Part A

1 point is earned for showing that $R(6) > 0$

Since $R(6) = 4.438 > 0$, the number of mosquitoes is increasing at $t = 6$.

0	1 ✓

The student response earns all of the following points:

1 point is earned for showing that $R(6) > 0$

Since $R(6) = 4.438 > 0$, the number of mosquitoes is increasing at $t = 6$.

Part B

1 point is earned for considering $R'(6)$

1 point is earned for the answer with reason

$R'(6) = -1.913$

Since $R'(6) < 0$, the number of mosquitoes is increasing at a decreasing rate at $t = 6$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for considering $R'(6)$

1 point is earned for the answer with reason

$R'(6) = -1.913$

Since $R'(6) < 0$, the number of mosquitoes is increasing at a decreasing rate at $t = 6$.

Part C

1 point is earned for the integral

1 point is earned for the answer

$$1000 + \int_0^{31} R(t)dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.

0	1	2 ✓

The student response earns all of the following points:

7-7b

1 point is earned for the integral

1 point is earned for the answer

$$1000+\int_0^{31} R(t)dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.

Part D

2 points are earned for absolute maximum value where 1 point is earned for the integral and 1 point is earned for the answer

2 points are earned for analysis where 1 point is earned for computing interior critical points and 1 point is earned for completing analysis

$R(t) = 0$ when $t = 0$, $t = 2.5\pi$, or $t = 7.5\pi$

$R(t) > 0$ on $0 < t < 2.5\pi$

$R(t) < 0$ on $2.5\pi < t < 7.5\pi$

$R(t) > 0$ on $7.5\pi < t < 31$

The absolute maximum number of mosquitoes occurs at $t = 2.5\pi$ or at $t = 31$.

$$1000+\int_0^{2.5\pi} R(t)dt = 1039.357,$$

There are 964 mosquitoes at $t = 31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

				✓
0	1	2	3	4

The student response earns all of the following points:

2 points are earned for absolute maximum value where 1 point is earned for the integral and 1 point is earned for the answer

2 points are earned for analysis where 1 point is earned for computing interior critical points and 1 point is earned for completing analysis

$R(t) = 0$ when $t = 0$, $t = 2.5\pi$, or $t = 7.5\pi$

$R(t) > 0$ on $0 < t < 2.5\pi$

$R(t) < 0$ on $2.5\pi < t < 7.5\pi$

$R(t) > 0$ on $7.5\pi < t < 31$

The absolute maximum number of mosquitoes occurs at $t = 2.5\pi$ or at $t = 31$.

$$1000+\int_0^{2.5\pi} R(t)dt = 1039.357,$$

There are 964 mosquitoes at $t = 31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

7-7b

75. Oil is being pumped continuously from a certain oil well at a rate proportional to the amount of oil left in the well; that is, $dy/dt = ky$, where y is the amount of oil left in the well at any time t. Initially there were 1,000,000 gallons of oil in the well, and 6 years later there were 500,000 gallons remaining. It will no longer be profitable to pump oil when there are fewer than 50,000 gallons remaining.

(a) Write an equation for y, the amount of oil remaining in the well at any time t.

(b) At what rate is the amount of oil in the well decreasing when there are 600,000 gallons of oil remaining?

(c) In order not to lose money, at what time t should oil no longer be pumped from the well?

Part A

2 points are earned for $y = e^{kt}$ **or** 1 point for each: $\ln|y| = kt + C$, $y = e^{kt} + C$

1 point is earned for $C = 10^6$

1 point is earned for $k = -\frac{\ln 2}{6}$

1 point is earned for answer

*Max 2/5 If y is not an exponential

$$\begin{array}{l} \frac{dy}{dt} = ky \\ y = Ce^{kt} \end{array} \quad \text{or} \quad \begin{cases} \frac{dy}{y} = k\,dt \\ \ln|y| = kt + C_1 \\ y = e^{kt + C_1} \end{cases}$$

$$t = 0 \Rightarrow C = 10^6,\ C_1 = \ln 10^6$$

$$\therefore y = 10^6 e^{kt}$$

$$t = 6 \Rightarrow \tfrac{1}{2} = e^{6k}$$

$$\therefore k = -\frac{\ln 2}{6}$$

$$y = 10^6 e^{\frac{-t}{6}\ln 2} = 10^6 \cdot 2^{\frac{-t}{6}}$$

0	1	2	3	4	5

The student response earns all of the following points:

2 points are earned for $y = Ce^{kt}$ **or** 1 point for each: $\ln|y| = kt + C$, $y = e^{kt} + C$

1 point is earned for $C = 10^6$

1 point is earned for $k = -\frac{\ln 2}{6}$

1 point is earned for answer

*Max 2/5 If y is not an exponential

7-7b

$$\frac{dy}{dt} = ky \quad \text{or} \quad \begin{cases} \frac{dy}{y} = k\,dt \\ \ln|y| = kt + C_1 \\ y = e^{kt+C_1} \end{cases}$$
$$y = Ce^{kt}$$

$t = 0 \Rightarrow C = 10^6,\ C_1 = \ln 10^6$

$\therefore y = 10^6 e^{kt}$

$t = 6 \Rightarrow \frac{1}{2} = e^{6k}$

$\therefore k = -\frac{\ln 2}{6}$

$y = 10^6 e^{\frac{-t}{6}\ln 2} = 10^6 \cdot 2^{\frac{-t}{6}}$

Part B

1 point is earned for answer consistent with part a.

$$\begin{aligned} \frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2 \end{aligned}$$

Decreasing at $10^5 \ln 2$ gal/year

0	1

The student response earns all of the following points:

1 point is earned for answer consistent with part a.

$$\begin{aligned} \frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2 \end{aligned}$$

Decreasing at $10^5 \ln 2$ gal/year

Part C

1 point is earned for solving $5 \cdot 10^4$

1 point is earned for *k.t* = -ln 20

1 point is earned for answer

$5 \cdot 10^4 = 10^6 e^{kt}$

$\therefore kt = -\ln 20 \ \left[= \ln \frac{1}{20}\right]$

$\therefore t = \frac{-\ln 20}{\frac{-\ln 2}{6}}$

$= 6\frac{\ln 20}{\ln 2} = 6 \log_2 20$

$6\frac{\ln 20}{\ln 2}$ years after starting

7-7b

0	1	2	3

The student response earns all of the following points:

1 point is earned for solving $5 \cdot 10^4$

1 point is earned for $k.t$ = -ln 20

1 point is earned for answer

$5 \cdot 10^4 = 10^6 e^{kt}$
$\therefore kt = -\ln 20 \; \left[= \ln \frac{1}{20}\right]$
$\therefore t = \frac{-\ln 20}{\frac{-\ln 2}{6}}$
$= 6 \frac{\ln 20}{\ln 2} = 6 \log_2 20$
$6 \frac{\ln 20}{\ln 2}$ years after starting

7-7b

76. A particle moves along the x-axis in such a way that its acceleration at time t for $t \geq 0$ is given by $a(t) = 4\cos(2t)$. At time $t = 0$, the velocity of the particle is $v(0) = 1$ and its position is $x(0) = 0$.

(a) Write an equation for the velocity $v(t)$ of the particle.

(b) Write an equation for the position $x(t)$ of the particle.

(c) For what values of t , $0 \leq t \leq \pi$, is the particle at rest?

Part A

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$\begin{aligned} v(t) &= \int 4\cos 2t\,dt \\ \therefore v(t) &= 2\sin 2t + C \\ v(0) &= 1 \Rightarrow C = 1 \\ \therefore v(t) &= 2\sin 2t + 1 \end{aligned}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$\begin{aligned} v(t) &= \int 4\cos 2t\,dt \\ \therefore v(t) &= 2\sin 2t + C \\ v(0) &= 1 \Rightarrow C = 1 \\ \therefore v(t) &= 2\sin 2t + 1 \end{aligned}$$

Part B

1 point is earned for integrand

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

1 point is earned for answer

$x(t) = \int 2\sin 2t + 1 \text{ dt}$

$\therefore x(t) = -\cos 2t + t + C$

$x(0) = 0 \Rightarrow C = 1$

$\therefore x(t) = -\cos 2t + t + 1$

0	1	2	3

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

7-7b

1 point is earned for answer

$x(t) = \int 2\sin 2t + 1 \text{ dt}$

$\therefore x(t) = -\cos 2t + t + C$

$x(0) = 0 \Rightarrow C = 1$

$\therefore x(t) = -\cos 2t + t + 1$

Part C

1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2\sin 2t + 1 = 0$$
$$\sin 2t = -\tfrac{1}{2}$$
$$\therefore 2t = \tfrac{7\pi}{6}, \tfrac{11\pi}{6}$$
$$\therefore t = \frac{7\pi}{12}, \frac{11\pi}{12}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2\sin 2t + 1 = 0$$
$$\sin 2t = -\tfrac{1}{2}$$
$$\therefore 2t = \tfrac{7\pi}{6}, \tfrac{11\pi}{6}$$
$$\therefore t = \frac{7\pi}{12}, \frac{11\pi}{12}$$

7-7b

77. A particle moves along the y-axis so that its velocity v at time $t \geq 0$ is given by $v(t) = 1 - \tan^{-1}(e^t)$. At time $t = 0$, the particle is at $y = -1$. (Note: $\tan^{-1} x = \arctan x$)

(a) Find the acceleration of the particle at time $t = 2$.

(b) Is the speed of the particle increasing or decreasing at time $t = 2$? Give a reason for your answer.

(c) Find the time $t \geq 0$ at which the particle reaches its highest point. Justify your answer.

(d) Find the position of the particle at time $t = 2$. Is the particle moving toward the origin or away from the origin at time $t = 2$? Justify your answer.

Part A

1 point is earned for the answer

$a(2) = v'(2) = -0.132$ or -0.133

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer

$a(2) = v'(2) = -0.132$ or -0.133

Part B

1 point is earned for the answer with reason

$v(2) = -0.436$

Speed is increasing since $a(2) < 0$ and $v(2) < 0$.

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer with reason

$v(2) = -0.436$

Speed is increasing since $a(2) < 0$ and $v(2) < 0$.

Part C

1 point is earned for: set $v(t) = 0$

1 point is earned for identifying $t = 0.433$ as a candidate

1 point is earned for justifying absolute maximum

$v(t) = 0$ when $\tan^{-1}(e^t) = 1$

$t = \ln(\tan(1)) = 0.433$ is the only critical value for y.

$v(t) > 0$ for $0 < t < \ln(\tan(1))$

$v(t) < 0$ for $t > \ln(\tan(1))$

$y(t)$ has an absolute maximum at $t = 0.433$.

0	1	2	3 ✓

7-7b

The student response earns all of the following points:

1 point is earned for: set $v(t) = 0$

1 point is earned for identifying $t = 0.433$ as a candidate

1 point is earned for justifying absolute maximum

$v(t) = 0$ when $\tan^{-1}\left(e^{t}\right) = 1$

$t = \ln(\tan(1)) = 0.433$ is the only critical value for y.

$v(t) > 0$ for $0 < t < \ln(\tan(1))$

$v(t) < 0$ for $t > \ln(\tan(1))$

$y(t)$ has an absolute maximum at $t = 0.433$.

Part D

1 point is earned for: $\int_0^2 v(t)dt$

1 point is earned for the handles initial condition

1 point is earned for the value of $y(2)$

1 point is earned for the answer with reason

$$y(2) = -1 + \int_0^2 v(t)dt = -1.360 \text{ or } -1.361$$

The particle is moving away from the origin since $v(2) < 0$ and $y(2) < 0$.

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for: $\int_0^2 v(t)dt$

1 point is earned for the handles initial condition

1 point is earned for the value of $y(2)$

1 point is earned for the answer with reason

$$y(2) = -1 + \int_0^2 v(t)dt = -1.360 \text{ or } -1.361$$

The particle is moving away from the origin since $v(2) < 0$ and $y(2) < 0$.

7-7b

78. A particle, initially at rest, moves along the x-axis so that its acceleration at any time $t \geq 0$ is given by $a(t) = 12t^2 - 4$. The position of the particle when $t = 1$ is $x(1) = 3$.

(a) Find the values of t for which t*he particle is at rest.*

(b) Write an expression for the position $x(t)$ of the particle at any time $t \geq 0$.

(c) Find the total distance traveled by the particle from $t = 0$ to $t = 2$.

Part A

2 points are earned for $v(t)$

1 point is **deducted** for leaving answer with c without solving for it

1 point is earned for setting $v(t) = 0$

1 point is earned for finding two non-negative roots

$$\begin{aligned} v(t) &= 4t^3 - 4t \\ v(t) &= 4t^3 - 4t = 0 \\ &4t\left(t^2 - 1\right) = 0 \\ \therefore t &= 0,\ t = 1 \end{aligned}$$

0	1	2	3	4 ✓

The student response earns all of the following points:

2 points are earned for $v(t)$

1 point is **deducted** for leaving answer with c without solving for it

1 point is earned for setting $v(t) = 0$

1 point is earned for finding two non-negative roots

$$\begin{aligned} v(t) &= 4t^3 - 4t \\ v(t) &= 4t^3 - 4t = 0 \\ &4t\left(t^2 - 1\right) = 0 \\ \therefore t &= 0,\ t = 1 \end{aligned}$$

Part B

2 point are earned for anti-derivative of $v(t)$

1 point is **deducted** for no constant of integration

1 point is earned for evaluating C

$$\begin{aligned} &x(t) = t^4 - 2t^2 + C \\ &3 = x(t) = 1^4 - 2 \cdot 1 + C \\ &3 = C - 1 \\ &4 = C \\ &x(t) = t^4 - 2t^2 + 4 \end{aligned}$$

0	1	2	3 ✓

The student response earns all of the following points:

7-7b

2 point are earned for anti-derivative of $v(t)$

1 point is **deducted** for no constant of integration

1 point is earned for evaluating C

$$x(t) = t^4 - 2t^2 + C$$
$$3 = x(t) = 1^4 - 2 \cdot 1 + C$$
$$3 = C - 1$$
$$4 = C$$
$$x(t) = t^4 - 2t^2 + 4$$

Part C

1 point is earned for considering resting points and computing distances

1 point is earned for determining total distance

1 point is **deducted** if resting point is NOT a turning point of student's work.

Note: 0/2 is earned if student never addresses a correct interior resting point for their $v(t)$

t	x(t)	distance
0	4	
		1
1	3	
		9
2	12	

Total Distance $\boxed{10}$

0	1	2

The student response earns all of the following points:

1 point is earned for considering resting points and computing distances

1 point is earned for determining total distance

1 point is **deducted** if resting point is NOT a turning point of student's work.

Note: 0/2 is earned if student never addresses a correct interior resting point for their $v(t)$

t	x(t)	distance
0	4	
		1
1	3	
		9
2	12	

Total Distance $\boxed{10}$

7-7b

79. An object moves along the x-axis with initial position $x(0) = 2$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \sin\left(\frac{\pi}{3}t\right)$.

(a) What is the acceleration of the object at time $t = 4$?

(b) Consider the following two statements.

Statement I: For $3 < t < 4.5$, the velocity of the object is decreasing.

Statement II: For $3 < t < 4.5$, the speed of the object is increasing.

Are either or both of these statements correct? For each statement provide a reason why it is correct or not correct.

(c) What is the total distance traveled by the object over the time interval $0 \leq t \leq 4$?

(d) What is the position of the object at time $t = 4$?

Part A

1 point is earned for the answer

$$a(4) = v'(4) = \frac{\pi}{3}\cos\left(\frac{4\pi}{3}\right)$$
$$= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524$$

0	1

The student response earns all of the following points:

1 point is earned for the answer

$$a(4) = v'(4) = \frac{\pi}{3}\cos\left(\frac{4\pi}{3}\right)$$
$$= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524$$

Part B

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

$$a(t) = v'(t) = \frac{\pi}{3}\cos\left(\frac{\pi}{3}t\right) < 0$$

Statement I is correct since $a(t) < 0$.

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

7-7b

$a(t) = v'(t) = \frac{\pi}{3}\cos\left(\frac{\pi}{3}t\right) < 0$

Statement I is correct since $a(t) < 0$.

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.

Part C

1 point is earned for limits of 0 and 4 on an integral of $v(t)$ or $|v\,(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for handle change of direction at student's turning point answer

1 point is earned for the answer

$\text{Distance} = \int_0^4 |v\,(t)|dt = 2.387$

OR

$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$
$x(0) = 2$
$x(4) = 2 + \frac{9}{2\pi} = 3.43239$
$v(t) = 0 \text{ When } t = 3$
$x(3) = \frac{6}{\pi} + 2 = 3.90986$
$|x\,(3) - x(0)| + |x\,(4) - x(3)| = \frac{15}{2\pi} = 2.387$

0	1	2	3

The student response earns all of the following points:

1 point is earned for limits of 0 and 4 on an integral of $v(t)$ or $|v\,(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for handles change of direction at student's turning point answer

1 point is earned for the answer

$\text{Distance} = \int_0^4 |v\,(t)|dt = 2.387$

OR

7-7b

$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$
$x(0) = 2$
$x(4) = 2 + \frac{9}{2\pi} = 3.43239$
$v(t) = 0$ When $t = 3$
$x(3) = \frac{6}{\pi} + 2 = 3.90986$
$|x(3) - x(0)| + |x(4) - x(3)| = \frac{15}{2\pi} = 2.387$

Part D

1 point is earned for integral

1 point is earned for the answer

OR

1 point is earned for $x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + C$

1 point is earned for the answer

$x(4) = x(0) + \int_0^4 v(t)dt = 3.432$

OR

$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$
$x(4) = 2 + \frac{9}{2\pi} = 3.432$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

OR

1 point is earned for $x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + C$

1 point is earned for the answer

$x(4) = x(0) + \int_0^4 v(t)dt = 3.432$

OR

$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$
$x(4) = 2 + \frac{9}{2\pi} = 3.432$

7-7b

80. A particle moves along the x-axis so that its velocity v at time t, for $0 \le t \le 5$, is given by $v(t) = \ln(t^2 - 3t + 3)$. The particle is at position $x = 8$ at time $t = 0$.

(a) Find the acceleration of the particle at time $t = 4$.

(b) Find all times t in the open interval $0 < t < 5$ at which the particle changes direction. During which time intervals, for $0 \le t \le 5$, does the particle travel to the left?

(c) Find the position of the particle at time $t = 2$.

(d) Find the average speed of the particle over the interval $0 \le t \le 2$.

Part A

1 point is earned for the answer

$a(4) = v'(4) = \frac{5}{7}$

0	1

The student response earns all of the following points:

1 point is earned for the answer

$a(4) = v'(4) = \frac{5}{7}$

Part B

1 point is earned for setting $v(t) = 0$

1 point is earned for the direction change at $t = 1, 2$

1 point is earned for the interval with reason

$v(t) = 0$

$$t^2 - 3t + 3 = 1$$
$$t^2 - 3t + 2 = 0$$

$(t - 2)(t - 1) = 0$

$t = 1, 2$

$v(t) > 0$ for $0 < t < 1$

$v(t) < 0$ for $1 < t < 2$

$v(t) > 0$ for $2 < t < 5$

The particle changes direction when $t = 1$ and $t = 2$.

The particle travels to the left when $1 < t < 2$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for setting $v(t) = 0$

1 point is earned for the direction change at $t = 1, 2$

1 point is earned for the interval with reason

7-7b

$v(t) = 0$

$$t^2 - 3t + 3 = 1$$
$$t^2 - 3t + 2 = 0$$

$(t - 2)(t - 1) = 0$

$t = 1, 2$

$v(t) > 0$ for $0 < t < 1$

$v(t) < 0$ for $1 < t < 2$

$v(t) > 0$ for $2 < t < 5$

The particle changes direction when $t = 1$ and $t = 2$.

The particle travels to the left when $1 < t < 2$.

Part C

1 point is earned for $\int_0^2 \ln\left(u^2 - 3u + 3\right) du$

1 point is earned if handles initial condition

1 point is earned for answer

$$s(t) = s(0) + \int_0^t \ln\left(u^2 - 3u + 3\right) du$$
$$s(2) = 8 + \int_0^2 \ln\left(u^2 - 3u + 3\right) du$$

$= 8.368$ or 8.369

0	1	2	3

The student response earns all of the following points:

1 point is earned for $\int_0^2 \ln\left(u^2 - 3u + 3\right) du$

1 point is earned if handles initial condition

1 point is earned for answer

$$s(t) = s(0) + \int_0^t \ln\left(u^2 - 3u + 3\right) du$$
$$s(2) = 8 + \int_0^2 \ln\left(u^2 - 3u + 3\right) du$$

$= 8.368$ or 8.369

Part D

1 point is earned for the integral

7-7b

1 point is earned for the answer

$$\frac{1}{2}\int_0^2 |v(t)|dt = 0.370 \text{ or } 0.371$$

0	1	2

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2}\int_0^2 |v(t)|dt = 0.370 \text{ or } 0.371$$

7-7b

81. At time t, $t \geq 0$, the volume of a sphere is increasing at a rate proportional to the reciprocal of its radius. At $t = 0$, the radius of the sphere is 1 and at $t = 15$, the radius is 2. (The volume V of a sphere with a radius r is $V = 4/3\pi r^3$.)

(a) Find the radius of the sphere as a function of t.

(b) At what time t will the volume of the sphere be 27 *times* its volume at $t = 0$?

Part A

1 point is earned for $\frac{dV}{dt} = \frac{k}{r}$

1 point is earned for $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

1 point is earned for separates variables

1 point is earned for integration (must have C)

1 point is earned for solving for C

1 point is earned for solving for k

1 point is earned for answer

$\frac{dV}{dt} = \frac{k}{r}$
$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
$\frac{k}{r} = 4\pi r^2 \frac{dr}{dt}$
$kdt = 4\pi r^3 dr \int kdt = \int 4\pi r^3 \frac{dr}{dt} dt$
$kt + C = \pi r^4$
At $t = 0,\ r = 1,$ so $C = \pi$
At $t = 15,\ r = 2,$ so
$15k + \pi = 16\pi, k = \pi$
$\pi r^4 = \pi t + \pi$
$r = \sqrt[4]{t+1}$

0	1	2	3	4	5	6	7

The student response earns all of the following points:

1 point is earned for $\frac{dV}{dt} = \frac{k}{r}$

1 point is earned for $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

1 point is earned for separates variables

1 point is earned for integration (must have C)

1 point is earned for solving for C

1 point is earned for solving for k

1 point is earned for answer

7-7b

$\frac{dV}{dt} = \frac{k}{r}$
$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
$\frac{k}{r} = 4\pi r^2 \frac{dr}{dt}$
$kdt = 4\pi r^3 dr \int kdt = \int 4\pi r^3 \frac{dr}{dt} dt$
$kt + C = \pi r^4$
At $t = 0,\ r = 1,$ so $C = \pi$
At $t = 15,\ r = 2,$ so
$15k + \pi = 16\pi, k = \pi$
$\pi r^4 = \pi t + \pi$
$r = \sqrt[4]{t+1}$

Part B

1 point is earned for $r = 3$

1 point is earned for answer

At $t = 0$, $r = 1$, so $V(0) = \frac{4}{3}\pi$

$27V(0) = 27\left(\frac{4}{3}\pi\right) = 36\pi$
$36\pi = \frac{4}{3}\pi r^3$
$r = 3$
$\sqrt[4]{t+1} = 3$
$t = 80$

0	1	2

The student response earns all of the following points:

1 point is earned for $r = 3$

1 point is earned for answer

At $t = 0$, $r = 1$, so $V(0) = \frac{4}{3}\pi$

$27V(0) = 27\left(\frac{4}{3}\pi\right) = 36\pi$
$36\pi = \frac{4}{3}\pi r^3$
$r = 3$
$\sqrt[4]{t+1} = 3$
$t = 80$

7-7b

82. The tide removes sand from Sandy Point Beach at a rate modeled by the function R, given by

$$R(t) = 2 + 5\sin\left(\frac{4\pi t}{25}\right).$$

A pumping station adds sand to the beach at a rate modeled by the function S, given by

$$S(t) = \frac{15t}{1+3t}.$$

Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \le t \le 6$. At time $t = 0$, the beach contains 2500 cubic yards of sand.

(a) How much sand will the tide remove from the beach during this 6-hour period? Indicate units of measure.

(b) Write an expression for $Y(t)$, the total number of cubic yards of sand on the beach at time t.

(c) Find the rate at which the total amount of sand on the beach is changing at time $t = 4$.

(d) For $0 \le t \le 6$, at what time t is the amount of sand on the beach a minimum? What is the minimum value? Justify your answers.

Part A

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t)dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t)dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$

Part B

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x))dx$$

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x))dx$$

7-7b

Part C

1 point is earned for the answer

$Y'(t) = S(t) - R(t)$
$Y'(4) = S(4) - R(4) = -1.908$ or -1.909 yd^3/hr

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer

$Y'(t) = S(t) - R(t)$
$Y'(4) = S(4) - R(4) = -1.908$ or -1.909 yd^3/hr

Part D

1 point is earned for setting $Y'(t) = 0$

1 point is earned for the critical t-value

1 point is earned for the answer with justification

$Y'(t) = 0$ when $S(t) - R(t) = 0$.

The only value in [0, 6] to satisfy $S(t) = R(t)$ is $a = 5.117865$.

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for setting $Y'(t) = 0$

1 point is earned for the critical t-value

1 point is earned for the answer with justification

$Y'(t) = 0$ when $S(t) - R(t) = 0$.

The only value in [0, 6] to satisfy $S(t) = R(t)$ is $a = 5.117865$.

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

7-7b

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.

7-7b

83. Let $v(t)$ be the velocity, in feet per second, of a skydiver at time t seconds, $t \geq 0$. After her parachute opens, her velocity satisfies the differential equation $\frac{dv}{dt} = -2v - 32$, with initial condition $v(0) = -50$.

(a) Use separation of variables to find an expression for v in terms of t, where t is measured in seconds.

(b) Terminal velocity is defined as $\lim_{t\to\infty} v(t)$. Find the terminal velocity of the skydiver to the nearest foot per second.

(c) It is safe to land when her speed is 20 feet per second. At what time t does she reach this speed?

Part A

Solution: Max 6/6

1 point is earned for correctly separating variables

$$\frac{dv}{dt} = -2v - 32 = -2(v + 16)$$

1 point is earned for the correct antiderivative of dv side

0/1 if not $\int \frac{dv}{av+b}$, $a \neq 0$

$$\frac{dv}{v+16} = -2dt$$

1 point is earned for the correct antiderivative of dt side

$$\ln|v + 16| = -2t + A$$

1 point is earned for the correct constant of integration

$$\left|v + 16\right| = e^{-2t+A} = e^A e^{-2t}$$

1 point is earned for correctly using initial condition $v(0) = -50$

$$v + 16 = Ce^{-2t}$$
$$-50 + 16 = Ce^0;\ \ C = -34$$

1 point is earned for correctly solving for $v(t)$

0/1 if not solving $\frac{dv}{av+b} = k\,dt$ where a, b, k nonzero

0/1 if no constant of integration

0/6 if variables not separated

$$v = -34e^{-2t} - 16$$

OR

Integrating Factor Solution: Max 5/6

1 point is earned for correct use of integrating factor

7-7b

$v'(t) = -2v(t) - 32$
$e^{2t}(v'(t) + 2v(t)) = -32e^{2t}$
$\frac{d}{dt}(v(t)e^{2t}) = -32e^{2t}$
$v(t)e^{2t} = \int -32e^{2t}dt = -16e^{2t} + C$
$v(t) = -16 + Ce^{-2t}$
$-50 = -16 + C;\;\; C = -34$

4 points are earned for the correct solution

$v(t) = -16 - 34e^{-2t}$

0	1	2	3	4	5	6

The student response earns all of the following points:

Solution: Max 6/6

1 point is earned for correctly separating variables

$$\frac{dv}{dt} = -2v - 32 = -2(v + 16)$$

1 point is earned for the correct antiderivative of *dv* side

0/1 if not $\int \frac{dv}{av + b}$, $a \neq 0$

$$\frac{dv}{v + 16} = -2dt$$

1 point is earned for the correct antiderivative of *dt* side

$\ln|v + 16| = -2t + A$

1 point is earned for the correct constant of integration

$\left|v + 16\right| = e^{-2t+A} = e^{A}e^{-2t}$

1 point is earned for correctly using initial condition $v(0) = -50$

$v + 16 = Ce^{-2t}$
$-50 + 16 = Ce^{0};\;\; C = -34$

1 point is earned for correctly solving for $v(t)$

0/1 if not solving $\frac{dv}{av + b} = k\,dt$ where a, b, k nonzero

0/1 if no constant of integration

0/6 if variables not separated

7-7b

$v = -34e^{-2t} - 16$

OR

Integrating Factor Solution: Max 5/6

1 point is earned for correct use of integrating factor

$v'(t) = -2v(t) - 32$
$e^{2t}(v'(t) + 2v(t)) = -32e^{2t}$
$\frac{d}{dt}(v(t)e^{2t}) = -32e^{2t}$
$v(t)e^{2t} = \int -32e^{2t}dt = -16e^{2t} + C$
$v(t) = -16 + Ce^{-2t}$
$-50 = -16 + C; \;\; C = -34$

4 points are earned for the correct solution

$v(t) = -16 - 34e^{-2t}$

Part B

1 point is earned for the correct limit value must be exponential $v(t)$ with finite limit

$\lim_{t\to\infty} v(t) = \lim_{t\to\infty}(-34e^{-2t} - 16) = -16$

0	1 ✓

The student response earns all of the following points:

1 point is earned for the correct limit value must be exponential $v(t)$ with finite limit

$\lim_{t\to\infty} v(t) = \lim_{t\to\infty}(-34e^{-2t} - 16) = -16$

Part C

1 point is earned for correctly setting $v(t) = -20$

$v(t) = -34e^{-2t} - 16 = -20$

1 point is earned for the correct solution must be exponential $v(t)$

$e^{-2t} = \frac{2}{17}; \;\; t = -\frac{1}{2}\ln\left(\frac{2}{17}\right) = 1.070$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for correctly setting $v(t) = -20$

7-7b

$v(t) = -34e^{-2t} - 16 = -20$

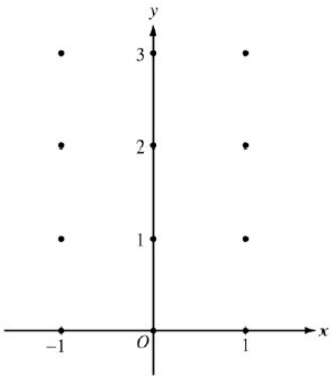
1 point is earned for the correct solution must be exponential $v(t)$

$e^{-2t} = \frac{2}{17}$; $t = -\frac{1}{2}\ln\left(\frac{2}{17}\right) = 1.070$

7-7b

84. Consider the differential equation $\frac{dy}{dx} = x^2(y-1)$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



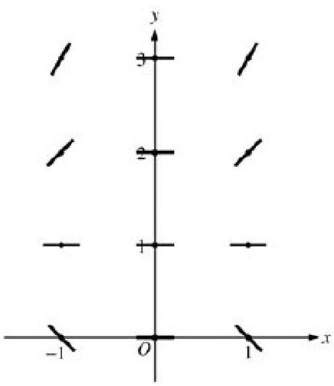
(b) While the slope field in part (a) is drawn at only twelve points, it is defined at every point in the xy-plane. Describe all points in the xy-plane for which the slopes are positive.

(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(0) = 3$.

Part A

1 point is earned for zero slope at each point (x, y) where $x = 0$ or $y = 1$

1 point is earned for positive slope at each point (x, y) where $x \neq 0$ or $y > 1$ and for negative slope at each point (x, y) where $x \neq 0$ or $y < 1$



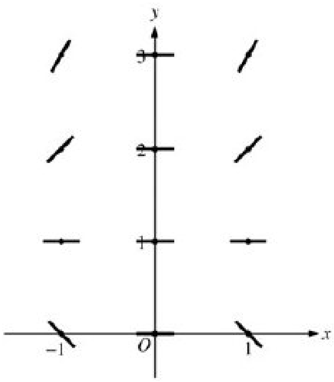
0	1	2

The student response earns all of the following points:

1 point is earned for zero slope at each point (x, y) where $x = 0$ or $y = 1$

1 point is earned for positive slope at each point (x, y) where $x \neq 0$ or $y > 1$ and for negative slope at each point (x, y) where $x \neq 0$ or $y < 1$

7-7b



Part B

1 point is earned for the description

Slopes are positive at points (x, y) where $x \neq 0$ and $y > 1$.

0	1

The student response earns all of the following points:

1 point is earned for the description

Slopes are positive at points (x, y) where $x \neq 0$ and $y > 1$.

Part C

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

0/1 if y is not exponential

$$\frac{1}{y-1}dy = x^2dx$$
$$\ln|y-1| = \frac{1}{3}x^3 + C$$
$$|y-1| = e^Ce^{\frac{1}{3}x^3}$$
$$y-1 = Ke^{\frac{1}{3}x^3}, K = \pm e^C$$
$$2 = Ke^0 = K$$
$$y = 1 + 2e^{\frac{1}{3}x^3}$$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separates variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

7-7b

1 point is earned for using initial condition

1 point is earned for solving for y

0/1 if y is not exponential

$\frac{1}{y-1}dy = x^2dx$
$\ln|y-1| = \frac{1}{3}x^3 + C$
$|y-1| = e^Ce^{\frac{1}{3}x^3}$
$y-1 = Ke^{\frac{1}{3}x^3}, K = \pm e^C$
$2 = Ke^0 = K$
$y = 1 + 2e^{\frac{1}{3}x^3}$

7-7b

85. A particle moves along the y-axis with velocity given by $v(t) = t \sin(t^2)$ for $t \geq 0$.

(a) In which direction (up or down) is the particle moving at time $t = 1.5$? Why?

(b) Find the acceleration of the particle at time $t = 1.5$. Is the velocity of the particle increasing at $t = 1.5$? Why or why not?

(c) Given that $y(t)$ is the position of the particle at time t and that $y(0) = 3$, find $y(2)$.

(d) Find the total distance traveled by the particle from $t = 0$ to $t = 2$.

Part A

1 point is earned for the answer and reason

$v(1.5) = 1.5\sin(1.5^2) = 1.167$

Up, because $v(1.5) > 0$

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer and reason

$v(1.5) = 1.5\sin(1.5^2) = 1.167$

Up, because $v(1.5) > 0$

Part B

1 point is earned for: $a(1.5)$

1 point is earned for the conclusion and reason

$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$
$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$

No; v is decreasing at 1.5 because $v'(1.5) < 0$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: $a(1.5)$

1 point is earned for the conclusion and reason

$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$
$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$

No; v is decreasing at 1.5 because $v'(1.5) < 0$

Part C

1 point is earned for: $y(t) = \int v(t)dt$

1 point is earned for: $y(t) = -\frac{1}{2}\cos t^2 + C$

7-7b

1 point is earn

$$\begin{aligned} y(t) &= \int v(t)dt \\ &= \int t\sin t^2 dt = -\frac{\cos t^2}{2} + C \\ y(0) &= 3 = -\tfrac{1}{2} + C \Rightarrow C = \tfrac{7}{2} \\ y(t) &= -\tfrac{1}{2}\cos t^2 + \tfrac{7}{2} \\ y(2) &= -\tfrac{1}{2}\cos 4 + \tfrac{7}{2} = 3.826 \text{ or } 3.827 \end{aligned}$$

ed for: $y(2)$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $y(t) = \int v(t)dt$

1 point is earned for: $y\,(t) = -\frac{1}{2}\cos t^2 + C$

1 point is earned for: $y(2)$

$$\begin{aligned} y(t) &= \int v(t)dt \\ &= \int t\sin t^2 dt = -\frac{\cos t^2}{2} + C \\ y(0) &= 3 = -\tfrac{1}{2} + C \Rightarrow C = \tfrac{7}{2} \\ y(t) &= -\tfrac{1}{2}\cos t^2 + \tfrac{7}{2} \\ y(2) &= -\tfrac{1}{2}\cos 4 + \tfrac{7}{2} = 3.826 \text{ or } 3.827 \end{aligned}$$

Part D

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or $|v\,(t)|$

or

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

0/1 if incorrect turning point

$$\text{distance} = \int_0^2 |v(t)|dt = 1.173$$

or

$$\begin{aligned} &v(t) = t\sin t^2 = 0 \\ &t = 0 \text{ or } t = \sqrt{\pi} \approx 1.772 \\ &y(0) = 3;\ y(\sqrt{\pi}) = 4;\ y(2) = 3.826 \text{ or } 3.827 \\ &\left[y(\sqrt{\pi}) - y(0)\right] + \left[y(\sqrt{\pi}) - y(2)\right] \\ &= 1.173 \text{ or } 1.174 \end{aligned}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or $|v\,(t)|$

7-7b

or

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

0/1 if incorrect turning point

distance $= \int_0^2 |v(t)| dt = 1.173$

or

$v(t) = t \sin t^2 = 0$

$t = 0$ or $t = \sqrt{\pi} \approx 1.772$

$y(0) = 3;\ y(\sqrt{\pi}) = 4;\ y(2) = 3.826$ or 3.827

$\left[y(\sqrt{\pi}) - y(0)\right] + \left[y(\sqrt{\pi}) - y(2)\right]$

$= 1.173$ or 1.174

7-7b

86. Water is pumped into an underground tank at a constant rate of 8 gallons per minute. Water leaks out of the tank at the rate of $\sqrt{t+1}$ gallons per minute, for $0 \leq t \leq 120$ minutes. At time $t = 0$, the tank contains 30 gallons of water.

(a) How many gallons of water leak out of the tank from time $t = 0$ to $t = 3$ minutes?

(b) How many gallons of water are in the tank at time $t = 3$ minutes?

(c) Write an expression for $A(t)$, the total number of gallons of water in the tank at time t.

(d) At what time t, for $0 \leq t \leq 120$, is the amount of water in the tank a maximum? Justify your answer.

Part A

Method 1:

2 points are earned for the definite integral

1: limits

1: integrand

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

Method 1: $\int_0^3 \sqrt{t+1}\, dt = \frac{2}{3}(t+1)^{3/2}\Big|_0^3 = \frac{14}{3}$

- or -

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1};\ L(t) = \tfrac{2}{3}(t+1)^{3/2} + C$$
$$L(0) = 0;\ C = -\tfrac{2}{3}$$
$$L(t) = \tfrac{2}{3}(t+1)^{3/2} - \tfrac{2}{3};\ \ L(3)\tfrac{14}{3}$$

0	1	2	3	4	5	6

The student response earns all of the following points:

Method 1:

2 points are earned for the definite integral

1: limits

1: integrand

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

7-7b

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

Method 1: $\int_0^3 \sqrt{t+1}\,dt = \frac{2}{3}(t+1)^{3/2}\Big|_0^3 = \frac{14}{3}$

- or -

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1};\ L(t) = \tfrac{2}{3}(t+1)^{3/2} + C$$
$$L(0) = 0;\ C = -\tfrac{2}{3}$$
$$L(t) = \tfrac{2}{3}(t+1)^{3/2} - \tfrac{2}{3};\ \ L(3)\frac{14}{3}$$

Part B

1 point is earned for the answer

$$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$$

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer

$$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$$

Part C

Method 1:

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1}\,dx$

- or -

Method 2:

1 point earned for antiderivative with C

1 point is earned for the answer

Method 1:

$$\begin{aligned} A(t) &= 30 + \int_0^t \left(8 - \sqrt{x+1}\right) dx \\ &= 30 + 8t - \int_0^t \sqrt{x+1}\,dx \end{aligned}$$

7-7b

- or -

Method 2:

$$\frac{dA}{dt} = 8 - \sqrt{t+1}$$
$$A(t) = 8t - \tfrac{2}{3}(t+1)^{3/2} + C$$
$$30 = 8(0) - \tfrac{2}{3}(0+1)^{3/2} + C;\;\; C = \frac{92}{3}$$
$$A(t) = 8t - \tfrac{2}{3}(t+1)^{3/2} + \frac{92}{3}$$

0	1	2	3	4

The student response earns all of the following points:

Method 1:

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1}\,dx$

- or -

Method 2:

1 point earned for antiderivative with C

1 point is earned for the answer

Method 1:

$$A(t) = 30 + \int_0^t \left(8 - \sqrt{x+1}\right)dx$$
$$= 30 + 8t - \int_0^t \sqrt{x+1}\,dx$$

- or -

Method 2:

$$\frac{dA}{dt} = 8 - \sqrt{t+1}$$
$$A(t) = 8t - \tfrac{2}{3}(t+1)^{3/2} + C$$
$$30 = 8(0) - \tfrac{2}{3}(0+1)^{3/2} + C;\;\; C = \frac{92}{3}$$
$$A(t) = 8t - \tfrac{2}{3}(t+1)^{3/2} + \frac{92}{3}$$

Part D

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$A'(t) = 8 - \sqrt{t+1} = 0$ when $t = 63$

7-7b

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$A'(t) = 8 - \sqrt{t+1} = 0$ when $t = 63$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.